

Reviewed Preprint

v1 • July 22, 2025

Not revised

Reviewed Preprint

v2 • March 19, 2026

Revised by authors

A quantitative in vivo CRISPR-imaging platform identifies regulators of hyperplastic and hypertrophic adipose morphology in zebrafish

✉ For correspondence:

james.minchin@ed.ac.uk

* Indicates equal contributions.

Competing interests:

No competing interests declared

Funding:

See [page 30](#)

Reviewing editor:

Michel Bagnat,

Duke University, United States

© 2025, Wafer et al. This article is

distributed under the terms of the

[Creative Commons Attribution](#)

[License](#), which permits unrestricted

use and redistribution provided that

the original author and source are

credited.

Rebecca Wafer*, Panna Tandon*, James EN Minchin ✉

Institute for Neuroscience and Cardiovascular Research, Edinburgh BioQuarter, College of Medicine and Veterinary Medicine, University of Edinburgh, Edinburgh, United Kingdom

eLife Assessment

In this manuscript, Wafer and Tandon et al. present a thoughtful and well-designed genetic screen for regulators of adipose remodeling using zebrafish as a model system. This work is **valuable** because it uncovers several genes associated with adipose tissue hyperplastic hypertrophic morphology and diet-induced remodeling that have considerable potential health impact. The rigorous phenotypic analyses and **compelling** evidence make this work a key resource for the field.

<https://doi.org/10.7554/eLife.107327.2.sa3>

Abstract

Adipose tissues exhibit a remarkable capacity to expand, regress, and remodel in response to energy status. The cellular mechanisms underlying adipose remodelling are central to metabolic health. Hypertrophic remodelling - characterised by the enlargement of existing adipocytes - is associated with insulin resistance, type 2 diabetes, and cardiovascular disease. In contrast, hyperplastic remodelling - in which new adipocytes are generated - is linked to improved metabolic outcomes. Despite its clinical importance, the regulation of hypertrophic and hyperplastic adipose morphology remains poorly understood. Here, we integrate human transcriptomic data with a quantitative CRISPR-imaging platform in zebrafish to identify regulators of adipose morphology. We developed an image-based phenotyping pipeline that captures lipid droplet size, number, and spatial patterning, and applied generalised additive modelling to quantify hyperplastic versus hypertrophic morphology signatures. Using this platform, we conducted an F0 CRISPR screen targeting 25 candidate genes and identified three that induced hypertrophic morphology (*txnipa*, *mmp14b* and *foxp1b*) and an additional candidate that altered total adiposity (*kazna*). For functional validation, we generated stable loss-of-function alleles for both zebrafish *foxp1* paralogues. Spatial analysis along the anterior-posterior axis revealed that *foxp1b* mutants display developmental hypertrophy but profoundly blunted adaptive responses to high-fat diet (~68% reduction across all spatial zones), while *foxp1a* mutants show normal baseline morphology but disrupted spatial patterning of diet-induced hypertrophy. Together, these findings establish a scalable CRISPR-imaging platform for in vivo genetic screening of adipose morphology, and reveal distinct roles for *Foxp1* paralogues in developmental patterning and adaptive responses to dietary challenge in adipose tissue.

Introduction

Adipose tissues exhibit a remarkable capacity to remodel and restructure themselves in response to the energetic status, the environment, physiological demands, or disease. Adipose remodelling involves alterations in both adipocyte number (addition or removal of adipocytes) and adipocyte

size (expansion or reduction in adipocyte size). Increases in adipocyte size (hypertrophy) can trigger hypoxia, necrosis, altered lipid flux, inflammation and insulin resistance (1–3). Accordingly, a hypertrophic adipose morphology – adipose characterised by few, large adipocytes – is associated with increased cardiometabolic disease (4–6). Contrastingly, a hyperplastic adipose morphology – characterised by many, small adipocytes – is associated with more metabolically beneficial outcomes (7,8). Indeed, reductions in the hyperplastic potential of mouse adipose is associated with decreased insulin sensitivity (9), and disruption of adipocyte progenitor differentiation, proliferation and renewal leads to adipose hypertrophy and ensuing metabolic dysfunction (10). Strikingly, adipose hypertrophy is increased and preadipocyte frequency is reduced in diabetic obese patients (relative to non-diabetic obese) (11), and genetic predisposition for Type 2 diabetes, but not obesity, is associated with an impaired ability to recruit new adipose cells to store excess lipids in the subcutaneous adipose tissue (12). Despite its clinical relevance, the genetic and cellular processes that determine hyperplastic versus hypertrophic adipose morphology remain poorly understood, and whether early developmental patterning of adipose morphology influences subsequent capacity for adaptive remodelling remains an open question.

The zebrafish provides a tractable and powerful *in vivo* model for studying adipose growth and remodelling dynamics. Zebrafish adipose tissue is morphologically similar to mammalian white adipose tissue, comprising large adipocytes with a single dominant cytoplasmic lipid droplet (13,14). Transcriptomic profiling further supports this conservation, with RNA-Seq data revealing strong molecular similarity between zebrafish and mammalian white adipose tissue (15). Functionally, zebrafish adipose depots respond dynamically to nutritional status – expanding following a high-fat diet and regressing following food restriction (16,17). Further, prolonged high-fat or overfeeding has been shown to induce diabetic phenotypes in zebrafish (18,19), reinforcing the model's relevance to metabolic disease. One of the key advantages of the zebrafish model is its optical accessibility and amenability to fluorescent imaging. Lipophilic dyes such as Nile Red can be used to label lipid droplets (LDs) *in vivo* (13,20), and LD-associated protein reporters now allow real-time visualisation of lipid droplet dynamics in living animals enabling rapid assessment of adiposity and LD morphology (21,22). Additionally, transgenic tools are increasingly available to label and manipulate zebrafish adipocytes (23,24). Importantly, zebrafish are amenable to high-throughput CRISPR mutagenesis, and recent studies have combined F0 screens with quantitative imaging to examine candidate obesity genes (25,26). However, existing approaches have focussed on total adiposity rather than cellular composition of adipose tissue. To date, no scalable platform exists to quantify hyperplastic versus hypertrophic adipose morphology *in vivo* – a gap that limits systematic genetic dissection of adipose growth properties.

Here, we integrate human transcriptomic data with a quantitative CRISPR-imaging platform in zebrafish to identify regulators of adipose morphology. We developed an image-based phenotyping pipeline that captures lipid droplet size, number, and spatial patterning within subcutaneous adipose, and applied generalised additive modelling to quantify hyperplastic versus hypertrophic morphology signatures. Using this platform, we screened 25 candidate genes and identified three that induced hypertrophic morphology (*txnipa*, *mmp14b* and *foxp1b*) and one that increased total adiposity (*kazna*). For functional validation, we focused on *Foxp1*, a transcription factor known to regulate stem and progenitor cell maintenance across multiple tissues. We generated stable loss-of-function alleles for both zebrafish *foxp1* paralogues and performed spatial analysis of lipid droplet dynamics along the anterior-posterior axis. This revealed that *foxp1b* mutants display developmental hypertrophy but show profoundly blunted adaptive responses to high-fat diet, while *foxp1a* mutants show normal baseline morphology but altered spatial patterning of diet-induced hypertrophy. These findings suggest distinct roles for *Foxp1* paralogues in developmental patterning versus adaptive remodelling of adipose tissue.

Methods

Identification of differentially expressed genes associated with small or large adipocytes in human subcutaneous adipose

Differentially expressed genes (DEGs) associated with large or small adipocytes in human SAT were obtained from Honecker et al., 2022. Briefly, 3,727 genes were significantly associated with adipocyte area by both categorical and continuous statistical models ($FDR < 0.05$) (27). 2,190 DEGs were associated with small adipocyte area, 1,537 DEGs associated with large adipocyte area (Suppl. Table 1). Expression within human WAT cell types for each of the candidate morphology genes was determined using data from (28) via the Single Cell Portal (https://singlecell.broadinstitute.org/single_cell). Scaled mean expression for each of the morphology DEGs was assessed within each of the 16 previously defined WAT cell clusters (adipose stem and progenitor cells (ASPCs), mesothelium, pericyte, adipocyte, macrophage, endothelial, LEC, monocyte, T cells, dendritic cells, smooth muscle cells, b cells, mast cells, natural killer cells, endometrium, and neutrophils). Of the 3,727 morphology genes, the following were removed: 410 genes without annotations in bulk RNA-Seq dataset, 250 genes not found in the Single Cell Portal dataset, and 87 genes not expressed in the 16 WAT cell types. Hierarchical clustering of the remaining 2,980 morphology genes based on WAT cell type expression was performed in Morpheus (<https://software.broadinstitute.org/morpheus>) using an average linkage method and one minus Pearson correlation metric (Suppl. Fig. 1). Morphology genes enriched in ASPCs were identified as either (i) belonging to the 75 morphology DEGs within the ASPC cluster, or (ii) one of the top 50 morphology DEGs with highest scaled mean expression in ASPCs (Suppl. Table 2). Three genes were present in both categories (*CXCL14*, *KAZN* and *SPARC*), resulting in 122 ASPC-enriched morphology DEGs (Suppl. Table 2). Ribosomal genes were excluded along with genes that did not code for proteins, resulting in 102 candidate morphology genes (Suppl. Table 2).

Prioritisation and ranking of candidate adipose morphology genes

Enrichment analysis on the 102 candidate genes described above was performed using GProfiler (<https://biit.cs.ut.ee/gprofiler/gost>), with the 3,727 original genes as a background gene set. Six enriched GO biological process gene sets were identified, comprising; three ‘developmental’ terms (GO:0048869, GO:0009888 & GO:0032502), one ‘differentiation’ term (GO:0030154), and two more generic ‘process’ terms (GO:0048869 & GO:0048523) (Suppl. Table 3). For each of the 102 morphology genes, beta coefficients between SAT expression levels and the 23 cardiometabolic traits from the METSIM study were clustered in Heatmapper (www.heatmapper.ca) using an average linkage clustering method and Pearson distance measurement method (Suppl. Fig. 2). Correlations between SAT expression and 23 cardiometabolic traits from 770 males as part of the METSIM study were obtained from (29). The following genes were not found in the METSIM data: *EIF4EBP3*, *PDF*, *SELENOM* and *SELENOS*. Candidate body mass index (BMI) and waist-to-hip ratio (WHR) genome-wide association study (GWAS) genes were identified by (i) downloading summary statistics for BMI and WHR from the GWAS Catalog (<https://www.ebi.ac.uk/gwas/>), (ii) identifying unique variants associated with the traits at genome-wide significance ($p < 5 \times 10^{-8}$), (iii) calculating proxy variants in high linkage disequilibrium in CEU population ($R^2 > 0.8$), (iv) calculating ‘LD blocks’ that encompass each lead and associated proxy variants and (v) identifying genes that overlap with the LD blocks. The DIOPT Ortholog Finder (https://www.flyrnai.org/cgi-bin/DRSC_orthologs.pl) was used to identify zebrafish orthologs of the candidate human morphology genes based on the weighted score. The ‘alignment & scores’ function in DIOPT was used to calculate protein similarity between human and zebrafish genes. scRNA-Seq scaled mean expression and % of cells expressing candidate genes from Yang Loureiro et al. (2023) were obtained from the Single Cell Portal.

Zebrafish husbandry and maintenance

Zebrafish were housed in the Queen's Medical Research Institute (QMRI) facility at the University of Edinburgh, UK. Standard husbandry was followed on a recirculating system and a 14h:10h light to dark cycle. Feeding regimen was as described in Tandon et al., 2025. The wild-type line used were WIKs maintained at the QMRI facility. Zebrafish experiments were conducted in accordance with the UK Animals (Scientific Procedures) Act 1986 under the project licence PP9112175.

F0 CRISPR screen in zebrafish

We followed general CRISPR screen methodology (30–32). Between three and five guide RNAs (gRNAs) were designed to target each gene using the UCSC Genome browser track ZebrafishGenomics (31) and CHOPCHOP software. gRNAs were selected based on off-target and efficiency scores, along with location in gene (targeting early exons, functional domains, and avoiding exons with lengths divisible by three). Pooled in vitro transcription of gRNAs was performed as per Wu et al (2018), and RNA was cleaned using ZymoClean columns (Zymo Research, Cat. # R1013). Prior to injection, gRNAs were incubated with Cas9 protein (New England Biolabs, Cat. # M0646T) and heated for 5 mins at 37°C. Mutagenesis of individual gRNAs was verified using T7E1 assays (New England Biolabs, Cat. # M0302S). gRNAs were injected into zebrafish embryos at the one-cell stage as per Wu et al 2018, along with a Fast Green dye to screen for successfully injected embryos. Identical injection mixes lacking the gRNAs (Cas9-only control) were used for control groups. 50 injected embryos were placed per petri dish and screened for survival at 5 days post fertilisation (dpf). At 5 dpf, injected larvae were placed in a small volume of system water in a 1L nursery tank without running water. From 10 dpf, a slow flow of system water was introduced. Larvae were raised at a density of 25 per 1L nursery tank, before being transferred to a 3L tank at 21 dpf. Survival was further assessed at 21 dpf. Feeding regimen was as described for standard husbandry. At 36–42 dpf, fish were euthanised and Nile Red staining performed as described previously (16).

Generation of *foxp1a* and *foxp1b* mutant zebrafish lines

foxp1a and *foxp1b* zebrafish mutant lines were made using standard CRISPR-Cas9 methods (34). For both genes we targeted the highly conserved DNA-binding Forkhead domain (FHD). The location and sequence of the CRISPR gRNAs were: *foxp1a* – 5'-GACGAGTGGAGAATGTGAAG-3' targeting the FHD in exon 14, and *foxp1b* – 5'-GATAACGAAGCATACGTGAA-3' targeting the FHD in exon 15. We generated two alleles – ed116 is a 13bp indel in the forkhead domain of *foxp1a* resulting in a frameshift and premature stop codon, and ed125 is a 4bp indel in the forkhead domain of *foxp1b* resulting in a frameshift and premature stop. Primers to amplify locus surrounding lesions for T7E1 assays or sequencing were: ed116 forward – 5'-GCCAGATTGGACTGG ATGTT-3', ed116 reverse – 5'-TTATTTCAGGCCATTCTGG-3', ed125 forward – 5'-TTCAGTTTCAGCTCC TTCCTTC-3', ed125 reverse – 5'-TGGAAGTCAAGCTACCAGCA-3'. To genotype ed116, primers were designed to recognise either the 13bp indel or the wild-type sequence. Sequences were: forward – 5'-GCCAGATTGGACTGGATGTT-3', reverse – 5'-TTATTTCAGGCCATTCTGG-3', wild-type – 5'-ACGAGTG GAGAATGTGAAGG-3', ed116 – 5'-AAACGGCCCCCTCGTACT-3'. For ed125, the following primers were used to amplify the locus: forward – 5'-TTCAGTTTCAGCTCCTTCCTTC-3', and reverse – 5'-TGGAAGTC AAGCTACCAGCA-3'. The PCR product was then digested with the restriction enzyme HpyCH4IV (New England Biolabs, Cat. # R0619S). Additionally, KASP genotyping assays were designed for the ed116 and ed125 alleles (KASP on demand, LGC Biosearch Technologies). Western blots were performed to assess effect of mutations on Foxp1 protein. Protein was isolated from the caudal fin – a pool of seven fin samples was placed on dry ice in RIPA buffer (Thermo, Cat. # 8990) containing protease inhibitor (Thermo, Cat. # A32953). Samples were homogenised with a TissueRupter, centrifuged for 15 mins at 4°C, lysate removed and added to LDS blue (Novex, Cat. # B0008) with reducing agent (Novex, B0009), heated to 95°C for 10 mins and the centrifuged for 1 min. Protein samples were run on 8% Bis-Tris gels (Invitrogen, NW00082BOX) and run together with protein ladder (Thermo Scientific, Cat. # 26619). Foxp1 antibody was PA5-26848 (Thermo Fisher) and β -actin antibody was (Sigma, A2228). Blood assays were from Biovision (Glucose – K606-100,

triacylglyceride – K622-100, cholesterol – K623-100) and performed as per the manufacturer's instructions with 1 ul of blood from an adult zebrafish used for the glucose and cholesterol assays, and 0.2 ul for the triacylglyceride assay. qRT-PCR was performed on a Roche LightCycler 480 using Luna reagents (New England Biolabs, Cat. # M3003L). cDNA was synthesised using Superscript IV (Invitrogen, Cat. # 18090010). Primers were: *foxp1a* forward 5'-GGCCACTTTGAGGAT GACTC-3', *foxp1a* reverse 5'-CCTCGCCACCTAAACTCAG-3', *foxp1b* forward 5'-CATTGGCTCCTCTTTTA CGC-3', *foxp1b* reverse 5'-ACAGCAACGGTAGTGACAGC-3', *bactin* forward 5'-GCCTCCGATCCAGACAGA GT-3' and *bactin* reverse 5'-TGACAGGATGCAGAAGGAGA-3'.

High-fat diet in zebrafish

A high-fat immersion diet was conducted on juvenile fish at 35 dpf. Genotyped fish were stained with Nile Red, imaged on a Leica M205 FCA stereoscope and recovered individually in a 12-well plate whilst the SL of each fish was measured. Based on SL, fish were placed into groups with equal average SL. Within groups, fish were randomly paired up and placed two fish per mesh insert in a 10L tank on the recirculating system, allowing pairs of fish to be tracked through the experiment. The location of each pair was randomly rotated daily to minimise tank location-effects. Fish assigned to the high-fat diet group were immersed in 5% chicken egg yolk for two hours daily over the course of 14 days (16). Control fish were placed in fresh system water for two hours daily. Both control and high-fat groups received an additional two feeds throughout the day.

Image-based adipose morphology profiling in zebrafish

For each Nile Red-stained juvenile zebrafish, we took two images on a Leica M205 FCA fluorescence stereomicroscope with a 1x objective: a whole animal image to allow us to measure total adiposity and standard length (SL), and a higher magnification image of the lateral subcutaneous adipose tissue (LSAT) along the zebrafish flank (hereafter referred to just as subcutaneous adipose or SAT). Images were always taken of the right side of the fish, as zebrafish adipose shows left-right asymmetry, with the right side containing the majority of pancreatic adipose. Images were first processed in Fiji/ImageJ and the 'Clear Outside' function was used to reduce image size and focus on the SAT area of interest or wider zebrafish. LD SAT images were either imported to Cellpose v2.2 and segmented using the 'cyto' model (33), or segmented in Napari using the SAMCell plugin and the samcell-cyto-large pre-trained model. Automated segmentations were assessed and manually adjusted to correct errors. Segmented masks and object outlines were imported to Fiji/ImageJ where xy coordinates for each segmented LD along with LD measurements were taken (including Feret's diameter for LD size). ImageJ macros used for the importation of Cellpose masks and measurement of xy coordinates are available at https://github.com/jeminchin/zebrafish_adipose_morphology_screen (34). Total adipose area and SAT area were segmented using Ilastik software as previously described (34). Standard length was measured using the line tool in Fiji/ImageJ.

Morphology value calculation

To quantify hyperplastic/hypertrophic adipose morphology we followed methodology from (35). Briefly, for a given total SAT area, we modelled the expected mean LD diameter using a generalised additive model (GAM):

$$Feret \sim s(\log(AreaSum))$$

where Feret is the mean LD Feret diameter, AreaSum is the total SAT area, and s denotes a penalised thin-plate regression spline fitted using the mgcv package in R. The GAM was fitted, for each gene, to all control (Cas9-only injected) fish pooled across experiments. Morphology value was calculated at the individual-fish level as the deviation of the observed mean Feret diameter from the GAM-predicted value. Positive deviations indicate a hypertrophic phenotype (larger LDs than expected for a given adipose depot size), while negative deviations indicate a hyperplastic phenotype (smaller, more numerous LDs).

Statistical framework for multi-experiment comparisons

Genes were tested across multiple independent injection experiments (typically 2–3 experiments per gene, see Fig. 5). Two genes (*aspa* and *lamb2*) had morphology data from only a single experiment and were analysed separately without batch correction. For genes with data from ≥ 2 experiments, we employed complementary statistical approaches to identify robust phenotypes while controlling for batch effects.

Kolmogorov-Smirnov tests

To assess distributional differences in morphology values between mutants and controls, we performed two-sample Kolmogorov-Smirnov (KS) tests:

Pooled KS test: Control and mutant data were pooled across all experiments. A GAM was fit to pooled control data, and KS tests compared the distribution of control residuals to mutant deviations from the control GAM.

Stratified KS test: To validate that effects were consistent within experiments (rather than driven by batch differences), we performed KS tests within each experiment separately, fitting experiment-specific GAMs to control data. Within-experiment p-values were combined using Fisher's method:

$$\chi^2 = -2 \sum_{i=1}^{\{k\}} \ln(p_i)$$

where k is the number of experiments. The combined statistic follows a χ^2 distribution with $2k$ degrees of freedom under the null hypothesis. Weighted mean effect sizes were calculated across experiments, weighting by sample size.

Linear mixed models

To quantify effect sizes while accounting for batch variation, we fit linear mixed models (LMMs) with experiment as a random intercept using the lme4 package.

Intercept model (uniform shift):

$$\log(\text{Feret}) \sim \log(\text{AreaSum}) + \text{Genotype} + (1|\text{Experiment})$$

This tests whether mutants have uniformly larger or smaller LDs across all adipose depot sizes.

Interaction model (size-dependent effect):

$$\log(\text{Feret}) \sim \log(\text{AreaSum}) \times \text{Genotype} + (1|\text{Experiment})$$

This tests whether the allometric scaling relationship between LD size and total adiposity differs between genotypes. A significant interaction indicates that the mutant phenotype magnitude depends on total adipose tissue size.

Effect sizes from the intercept model were back-transformed to percentage changes in Feret diameter: % change = $(\exp(\beta) - 1) \times 100$, where β is the genotype coefficient. 95% confidence intervals were calculated as $(\exp(\beta \pm 1.96 \times \text{SE}) - 1) \times 100$.

Multiple testing correction

Benjamini-Hochberg false discovery rate (FDR) correction was applied separately to: (i) pooled KS tests, (ii) stratified KS tests (Fisher's combined p-values), (iii) LMM intercept p-values, and (iv) LMM slope p-values. Adjusted p-values < 0.05 were considered significant.

Robustness criteria

Genes were classified as robust hits if they reached significance (adjusted $p < 0.05$) in both the stratified KS test and LMM intercept test. This dual criterion ensures that: (i) the distributional effect is consistent within experiments (not a batch artifact), and (ii) the mean shift is statistically significant after accounting for repeated measures.

Variance heterogeneity analysis

To test whether mutants showed altered phenotypic variance beyond mean shifts, we performed F-tests comparing the variance of morphology values between genotypes. For each gene, variance ratio was calculated as $\sigma^2(\text{mutant})/\sigma^2(\text{control})$. Higher-order distributional moments were calculated using the moments package in R: skewness (asymmetry of distribution) and excess kurtosis (tail weight relative to normal distribution). The percentage of observations exceeding 2 standard deviations from the mean (“tail percentage”) was also calculated. P-values from F-tests were corrected using Benjamini-Hochberg FDR.

LMM robustness verification

To verify that LMM results were not artifacts of violated model assumptions, we performed sensitivity analyses using three alternative approaches:

Heteroscedastic LMM: Linear mixed models allowing different residual variances per genotype were fit using the nlme package, with variance structure specified as `weights = varIdent(form = ~1|genotype)`.

Permutation test: Genotype labels were permuted 1,000 times within each experiment to generate a null distribution of genotype effects. Permutation p-values were calculated as the proportion of permuted effects exceeding the observed effect.

Welch’s t-test: Two-sample t-tests with Welch-Satterthwaite degrees of freedom correction (not assuming equal variance) were performed on morphology residuals.

Results were classified by consensus: “ALL SIG” if all methods yielded $p < 0.05$, “ALL NS” if no methods were significant, or “MIXED” if methods disagreed. Only genes with consistent significance across all sensitivity methods were considered to have robust LMM estimates.

Single-experiment gene analysis

For *aspa* and *lamb2* (single experiment each for morphology), batch correction was not possible. These genes were analysed using standard linear regression:

$$\log(\text{Feret}) \sim \log(\text{AreaSum}) + \text{Genotype}$$

KS tests were also performed. Results are reported without FDR correction and should be interpreted as preliminary findings requiring validation.

Software and code availability

All analyses were performed in R (version 4.5.1) using the following packages: mgcv for GAM fitting, lme4 and lmerTest for linear mixed models, nlme for heteroscedastic mixed models, and moments for distributional statistics. Code and data are available as a reusable pipeline at https://github.com/jeminchin/zebrafish_adipose_morphology_screen.

Results

Identification of 102 candidate human adipose morphology genes enriched in adipocyte stem and progenitor cells

To identify candidate regulators of hyperplastic/hypertrophic adipose morphology, we leveraged previously published bulk RNA-Seq data that reported 3,727 differentially expressed genes (DEGs) in human subcutaneous adipose tissue (SAT) characterised by either large or small adipocytes (Fig. 1; Suppl. Table 1) (27). We reasoned these DEGs may contribute to defining hypertrophic or hyperplastic adipose phenotypes. As the RNA-Seq was performed on whole SAT - comprising a heterogeneous mix of cell types - we next asked which adipose-resident cell populations preferentially express these morphology-associated genes. To address this, we clustered the 3,727 candidate genes based on their relative expression across 16 human white adipose tissue (WAT)

cell types, using published single-cell transcriptomic data (Fig. 1 [↗](#); Suppl. Fig. 1 [↗](#)) (28). As expected, the largest subset of morphology DEGs ($n = 730$) was enriched in mature adipocytes, consistent with central roles for adipocytes in defining overall adipose morphology (Suppl. Fig. 1 [↗](#)). Notably, we also identified 102 DEGs enriched in adipose stem and progenitor cells (ASPCs) (Fig. 1 [↗](#); Suppl. Fig. 1 [↗](#); Suppl. Table 2 [↗](#)). Given prior evidence that ASPC abundance and differentiation rate can influence adipose tissue expansion and morphology (36,37), we hypothesised that these genes may function within ASPCs to regulate or establish adipose morphology.

Analysis of candidate human adipose morphology genes using GWAS and cardiometabolic trait associations

Prior to functional analysis, we first explored genetic and cardiometabolic features of the 102 candidate genes. As an initial step, we examined overlap with BMI- and WHR-associated GWAS genes, and found that 16 of the candidate morphology genes were also BMI GWAS genes (*TNRC6B*, *LRMDA*, *PRRX1*, *TSHZ2*, *PTPRD*, *PARD3*, *ADH1B*, *TBX15*, *MAGI2*, *RTL8A*, *CAST*, *BNC2*, *RERE*, *PTPRG*, *ADAMTSL1* and *C1QBP*) and one gene, *HOXC4*, overlapped with WHR-associated loci; however, Fisher's exact test indicated these overlaps were not significantly enriched (Suppl. Fig. 2 [↗](#)). We next clustered candidate genes based on their SAT expression correlations with 23 cardiometabolic traits from the METSIM study (29). This revealed two distinct gene–trait signatures: one associated with a “healthy” metabolic profile - characterised by higher hip circumference, fat-free mass, HDL-C, Muscle Insulin Sensitivity Index (MISI) and adiponectin levels, alongside lower BMI, WHR, total fatty acids, triacylglycerides, Insulin and HOMA-B - and another with an “unhealthy” profile, showing the opposite trait pattern (Suppl. Fig. 2 [↗](#)). Notably, genes associated with both hypertrophic and hyperplastic adipose morphology were found across both signatures (Suppl. Fig. 2 [↗](#)), suggesting that these morphological phenotypes are not strictly aligned with traditional markers of metabolic health.

Enrichment of cell differentiation and developmental genes among adipose morphology candidates

To explore functional themes within our gene set, we performed enrichment analysis to identify overrepresented biological processes. This analysis revealed that 54 of the 102 ASPC-enriched genes were significantly associated with Gene Ontology (GO) terms related to cell differentiation (GO:0030154) and tissue development (GO:0048869, GO:0009888, GO:0032502) (Figs. 1 [↗](#), 2A,B [↗](#) & Suppl. Table 3 [↗](#)). Reasoning that they may play key roles in ASPC differentiation and/or the patterning of adipose tissue, we prioritised this subset of 54 genes for further analysis. To refine this list, we applied two complementary strategies. First, we ranked genes by their sequence conservation between humans and zebrafish (Fig. 2D [↗](#)). Second, we assessed scRNA-Seq expression dynamics during progenitor differentiation to adipocytes (38) (Figs. 2D–F [↗](#)). From this, we shortlisted 25 highly conserved genes that were predominantly expressed in either progenitors from undifferentiated samples, or within multipotent progenitors that persist following differentiation (*MGP*+ SWAT cells) (Figs. 2D–F [↗](#) & Table 1 [↗](#)). This final set included genes with established roles in adipocyte progenitors – such as *PRRX1* (H. Liu et al., 2022), *TBX15* (Pan et al., 2021), and *IRX1* (Han, 2022) - though their specific contributions to adipose morphology have not yet been characterised.

An image-based pipeline to quantify lipid droplet size, number and spatial patterning in zebrafish subcutaneous adipose

To investigate the function of the candidate morphology genes, we developed an image-based phenotyping pipeline in zebrafish. Building on previous studies where Nile Red fluorescence was used to quantify whole-animal adiposity and perform targeted CRISPR screens for regulators of total body fat (16,26), we refined this approach to capture not only global adiposity data but also high-resolution images of Nile Red-positive LDs. This enabled detailed quantification of LD size,

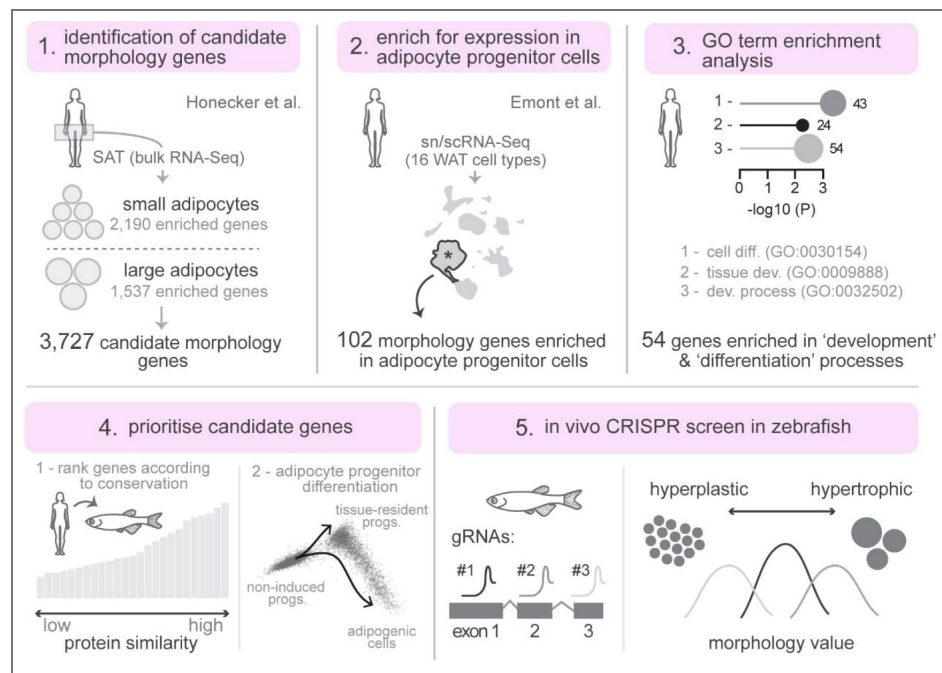


Figure 1. Experimental workflow.

(1) Identification of candidate morphology genes from bulk RNA-Seq of human subcutaneous adipose tissue (SAT), comparing genes enriched in small versus large adipocytes (Honecker et al.). (2) Filtering for expression in adipose stem and progenitor cells using sn/scRNA-Seq of 16 white adipose tissue (WAT) cell types (Emont et al.). (3) GO term enrichment analysis to focus on genes involved in development and differentiation processes. (4) Prioritisation of candidate genes based on protein conservation to zebrafish and expression dynamics during adipocyte progenitor differentiation. (5) In vivo F0 CRISPR screen in zebrafish using multiple gRNAs per gene to quantify lipid droplet morphology (hyperplastic versus hypertrophic).

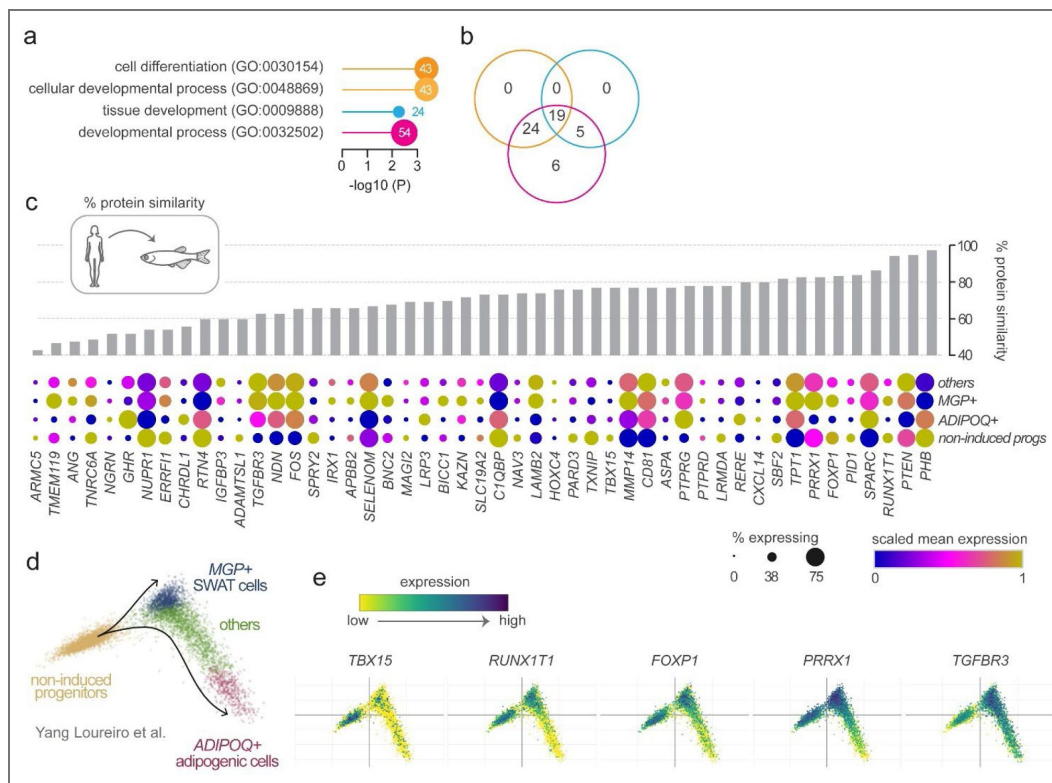


Figure 2. Prioritisation of candidate adipose morphology genes based on conservation to zebrafish and expression dynamics during progenitor differentiation.

A. GO term enrichment analysis of the 102 candidate morphology genes enriched in adipose stem and progenitor cells. Four GO terms related to development and differentiation were significantly enriched, encompassing 54 genes. **B.** Venn diagram showing gene overlap between the enriched GO terms. Cell differentiation (GO:0030154) and cellular developmental process (GO:0048869) had 100% overlap. **C.** Protein conservation and expression during adipocyte progenitor differentiation for each candidate gene. Grey bars denote percentage protein similarity between human and zebrafish orthologues. Dot plots show scaled mean expression (colour) and percentage of cells expressing each gene (dot size) across four cell populations identified during in vitro progenitor differentiation: non-induced progenitors, MGP+ SWAT cells, ADIPOQ+ adipogenic cells and others (Yang Loureiro et al., 2023). Genes are ordered by protein similarity. **D.** PCA projection of single cells coloured by cluster designation from Yang Loureiro et al., 2023. **E.** Expression of selected candidate morphology genes projected onto the PCA embedding, illustrating enrichment during specific stages of adipocyte differentiation.

Gene	Reps	n (ctrl/mut)	KS test			LMM		Direction	Power (d = 0.8)
			Effect (μ m)	Pooled (FDR adj. p value)	Stratified (FDR adj. p value)	Effect (%)	FDR adj. p value		

Table 1. Zebrafish gene targets and adipose morphology statistics.

Adjusted p values after Benjamini-Hochberg FDR correction for 21 statistical tests. KS = Kolmogorov-Smirnov test. LMM = linear mixed model with experiment as random effect. Values in bold indicate statistical significance. Gene names in bold indicate robust phenotypes (significant in both stratified KS and LMM). Stratified KS tests were performed within each experiment and combined using Fisher's method. Genes with only one replicate (*aspa*, *lamb2*) or which showed lethality (*phb*, *rerea*) are not included in this table.

<i>txnipa</i>	3	45/30	+4.6	0.066	1.41e-04	+19.9%	1.63e-04	Hypertrophic	92%
<i>mmp14b</i>	2	34/30	+5.4	0.002	0.005	+15.8%	1.93e-04	Hypertrophic	88%
<i>foxp1b</i>	2	42/45	+8.8	0.064	0.019	+17.0%	0.049	Hypertrophic	96%
<i>ptprdb</i>	2	35/24	+4.0	0.080	0.057	+18.5%	0.087	Hypertrophic	84%
<i>cxcl14</i>	2	27/24	-4.4	0.522	0.077	-8.7%	0.348	Hyperplastic	80%
<i>prrx1b</i>	3	65/60	-3.5	0.178	0.077	-5.9%	0.358	Hyperplastic	99%
<i>ptenb</i>	2	27/25	-4.7	0.054	0.077	-7.9%	0.411	Hyperplastic	81%
<i>pid1</i>	2	27/26	-2.7	0.869	0.204	-6.5%	0.565	Hyperplastic	81%
<i>nav3</i>	3	49/42	-2.4	0.194	0.562	-6.1%	0.650	Hyperplastic	96%
<i>sdk1a</i>	2	29/25	+0.6	0.869	0.482	+7.6%	0.694	Hypertrophic	82%
<i>runx1t1</i>	2	29/22	+1.1	0.706	0.795	+7.6%	0.736	Hypertrophic	79%
<i>sparc</i>	2	24/23	-2.1	0.286	0.553	-7.1%	0.736	Hyperplastic	77%
<i>kazna</i>	2	36/23	-3.1	0.706	0.148	-4.7%	0.759	Hyperplastic	84%
<i>tmem115</i>	2	27/29	-3.1	0.488	0.019	-3.8%	0.759	Hyperplastic	84%
<i>cd81a</i>	2	24/26	+0.0	0.862	0.064	-5.0%	0.759	Hypertrophic	79%
<i>lrmda</i>	2	27/22	-0.4	0.527	0.077	-2.7%	0.778	Hyperplastic	78%
<i>tbx15</i>	3	37/26	+3.4	0.706	0.019	+3.7%	0.778	Hypertrophic	87%
<i>cacnb4a</i>	2	29/27	+0.9	0.706	0.852	+2.6%	0.804	Hypertrophic	84%
<i>pard3ab</i>	2	36/32	-1.5	0.869	0.750	+2.1%	0.804	Hypertrophic	90%
<i>ptprga</i>	2	35/35	-1.4	0.269	0.077	-1.6%	0.804	Hyperplastic	91%
<i>srpx</i>	2	36/20	-5.7	0.155	0.204	-0.9%	0.836	Hyperplastic	80%

Table 1. (continued)

number and spatial patterning within zebrafish SAT (Fig. 3A-D). High-magnification images were acquired with a pixel size between 1-2 μm^2 allowing individual LDs to be clearly resolved within zebrafish SAT (Fig. 3A-D). Although very small LDs within multilocular adipocytes fall below our detection threshold, this resolution reliably captures the larger LDs characteristic of mature adipocytes (Fig. 3A-D). Using this pipeline, we characterised SAT development in 107 wild-type zebrafish from three independent clutches (Fig. 1E-G). Representative LD masks illustrating the morphological progression of SAT expansion are shown in Fig. 3E. Consistent with previous observations, LDs are first deposited at the anterior end of the adipose depot and new LDs are added in both posterior and dorsoventral directions (Fig. 3E). To spatially track this progression, we annotated the most anterior LD and defined successive 200 μm strata extending posteriorly from this reference point (Fig. 3E). This stratification enabled comparisons of LD morphology and number at anatomically and developmentally equivalent positions across individuals. As zebrafish grew from 7 to 12 mm standard length (SL), we observed limited increases in LD number within the anterior-most stratum (stratum 1) (Fig. 3E & F). Instead, new LDs accumulated in more posterior regions, particularly in strata 3 and 4, which exhibited the greatest LD number and overall growth (Figs. 3E & F). Concurrently, LDs increased in size, reaching an average diameter of $\sim 65 \mu\text{m}$ by 11–12 mm SL (Fig. 3E & G). Notably, posterior strata - containing more recently formed LDs - appeared to be in the process of enlarging toward this consistent size benchmark (Fig. 3E & G). Together, these findings show that during these relatively young juvenile stages, LDs expand rapidly in size but ultimately stabilise at an average diameter of $\sim 65 \mu\text{m}$, reflecting a potential upper limit of LD growth at this developmental stage.

Quantification of adipose morphology in juvenile zebrafish

As LD size, number and spatial dynamics can be robustly captured in juvenile zebrafish, we next developed methodology to quantify adipose morphology – specifically hyperplastic versus hypertrophic patterns – in a format scalable for CRISPR-based screening. Our approach was conceptually based on previously described methods for assessing human adipose morphology (35). To establish a baseline, we quantified SAT LD number and size within 194 wild-type fish (including the 107 animals analysed in Fig. 3). We observed a strong curvilinear relationship between mean LD diameter and the total depot size ($R^2 = 0.85$) (Fig. 4A). This relationship also held true for other LD size metrics (including area and perimeter), further suggesting that as the zebrafish SAT expands, LDs initially grow rapidly in size but then plateau as they reach a steady-state size and hypertrophic growth slows relative to the overall size of the depot. Following the strategy of Arner et al (2010), we calculated a ‘morphology value’ for each individual based on the deviation of each fish from the fitted curve (Fig. 4B). Morphology values were normally distributed around zero, with approximately equal proportions of individuals exhibiting hyperplastic (negative deviation) or hypertrophic (positive deviation) SAT (Fig. 4B). Notably, morphology values were inversely correlated with LD number ($p = 3.9 \times 10^{-6}$), such that hypertrophic individuals had fewer, larger LDs while hyperplastic individuals had more numerous, smaller LDs (Fig. 4C). Together, these data show that morphology values can be applied to zebrafish SAT and used to quantify hyperplastic/hypertrophic patterning.

Morphology values can be used to assess differences in F0 zebrafish CRISPR mutants

We next tested whether this metric could detect known regulators of adipose morphology. As a positive control, we targeted *growth hormone 1* (*gh1*) using established F0 CRISPR mutant methodology. Consistent with previous findings that *gh1* mutants display hypertrophic SAT LDs (39), *gh1* F0 CRISPR mutants displayed significantly larger LDs and an upward shift in morphology values ($p = 0.04$) (Suppl. Fig. 3). Notably, this effect was detected in only nine *gh1*-targeted fish, corresponding to a large effect size (Cohen’s $d = 1.06$) (Suppl. Fig. 3). Power analysis based on this dataset indicated that similar effects could be detected with 80% power using as few as 11 fish. Moreover, detecting medium-sized effects (Cohen’s $d = 0.6$ -0.7) would require only 23-31 mutant

Figure 3. Spatial dynamics of subcutaneous adipose growth in zebrafish.

A. Nile Red-stained zebrafish at 11.6 mm standard length (SL). Black signal denotes Nile Red-positive neutral lipid within adipose tissue. Magenta dashed box indicates the region shown in (B). **B.** Higher magnification of the subcutaneous adipose depot. Magenta outlines highlight segmented lipid droplets (LDs). Dashed line indicates the operculum. **C.** Zoomed view of individual subcutaneous adipose LDs. White dots are melanosome pigment granules. **D.** LD segmentation masks (magenta outlines) corresponding to (C). **E.** Segmented subcutaneous adipose LDs from four representative zebrafish of increasing body size, colour-coded by LD diameter. Fish sizes are shown in mm SL. Dashed lines demarcate strata defined at 200 μm intervals from the most anterior LD. **F.** Number of LDs per stratum as a function of zebrafish body size. Fish were grouped into five SL categories (colour-coded). Lines represent fitted curves; dots represent individual fish. **G.** Mean LD diameter per stratum as a function of zebrafish body size ($n = 107$ fish). Lines and colour coding as in (F).

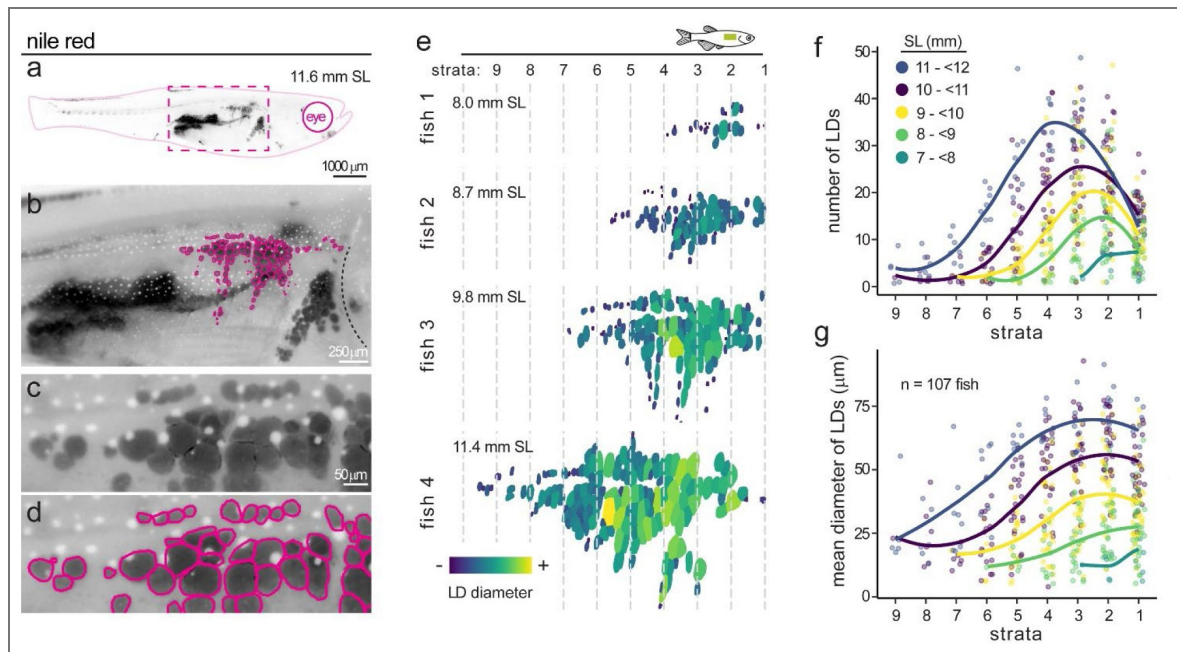
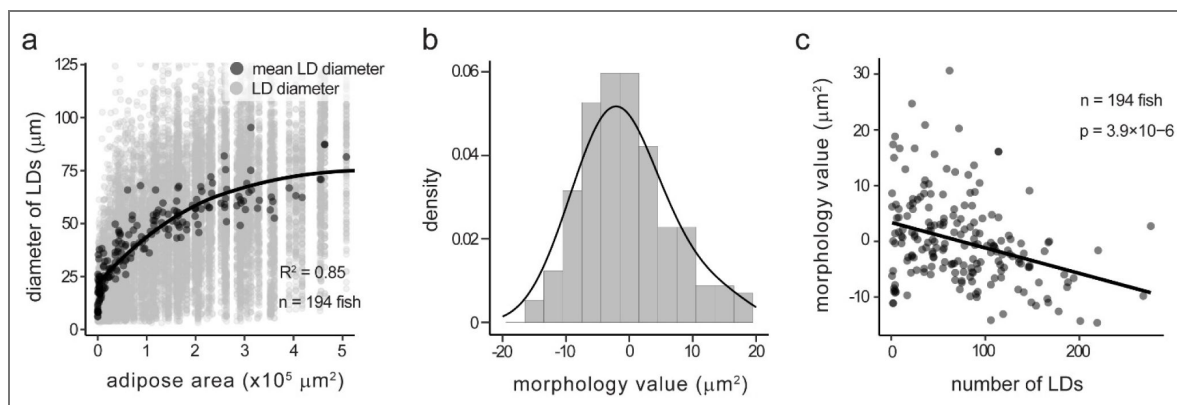


Figure 4. Derivation of the adipose morphology value from the relationship between lipid droplet size and adipose depot size.

A. Diameter of subcutaneous adipose LDs plotted against total depot area. Light grey dots represent individual LDs; dark grey dots represent the mean LD diameter per fish. Black line was fitted using a generalised additive model (GAM; $R^2 = 0.85$, $n = 194$ fish). **B.** Distribution of morphology values, calculated as the residual deviation of each fish's mean LD diameter from the GAM fit in (A). Positive values indicate larger LDs than expected (hypertrophic); negative values indicate smaller LDs than expected (hyperplastic). Black line shows a normal density fit. **C.** Morphology value plotted against number of LDs per fish, showing an inverse relationship (linear model, $p = 3.9 \times 10^{-6}$, $n = 194$ fish). Fish with hypertrophic morphology tend to have fewer, larger LDs, while fish with hyperplastic morphology have more numerous, smaller LDs.



fish. Together, these results demonstrate that image-based profiling of adipose morphology is a sensitive and scalable approach for in vivo genetic screens aimed at identifying regulators of hyperplastic and hypertrophic adipose growth.

A targeted CRISPR screen identified three genes that induce hypertrophic morphology in subcutaneous adipose

Following validation of our morphology quantification approach using *gh1*, we applied the same F0 CRISPR methodology and image-based pipeline to functionally assess all 25 conserved candidate genes identified from human adipose tissue datasets. In total, we screened 1,371 juvenile zebrafish comprising 738 Cas9-only controls and 633 gene-targeted mutants across 2-3 independent experimental replicates per gene (Fig. 5A-C & Table 1). Two targets, *phb* and *rerea*, were lethal prior to 5 dpf and therefore excluded from morphological analysis. Of the remaining 23 genes, none showed significant effects on standard length after controlling for batch effects, indicating that CRISPR targeting did not cause generalised growth defects (Fig. 5D). One gene (*kazna*) showed significantly increased total adiposity (Fig. 5E).

To quantify SAT morphology and classify hyperplastic or hypertrophic profiles, we applied two complementary statistical approaches. First, we calculated morphology values for each fish and compared mutant versus control morphology value distributions using the Kolmogorov-Smirnov (KS) test. This non-parametric test does not assume a specific distribution and is sensitive to shape-based distributional changes. To ensure effects were consistent across experimental batches rather than driven by between-experiment variability, we performed both pooled KS tests (combining data across all experiments) and stratified KS tests (testing within each experiment separately, then combining p-values using Fisher's method). Second, we applied linear mixed-models (LMMs) to evaluate the relationship between LD size and total SAT area, whilst accounting for replicate-specific variability. The LMMs allowed us to test whether LDs scaled normally with tissue size (slope) or differed in size at any given adipose area (intercept). After correcting for multiple tests using the Benjamini-Hochberg procedure, we identified three genes with robust hypertrophic morphology phenotypes validated by both stratified KS tests (which control for batch effects by testing within each experiment) and linear mixed models: *txnipa* (+19.9% increase in LD diameter, stratified KS adj.p = 1.41×10^{-4} , LMM adj.p = 1.63×10^{-4}), *mmp14b* (+15.8%, stratified KS adj.p = 0.005, LMM adj.p = 1.93×10^{-4}), and *foxp1b* (+17.0%, stratified KS adj.p = 0.019, LMM adj.p = 0.049) (Table 1 & Fig. 5). Two genes (*tmem115* and *tbx15*) showed significant distributional shifts in stratified KS tests (adj.p = 0.019 for both) but non-significant LMM intercept effects, suggesting altered variance or size-dependent phenotypes rather than uniform shifts in LD size (Table 1). Indeed, *tbx15* mutants displayed a significantly steeper size-scaling slope (+0.107, adj.p = 0.026), indicating that any hypertrophic tendency is more pronounced in larger fish (Table 1). Three additional genes showed nominally significant effects in pooled KS tests prior to FDR correction but were not validated by stratified within-experiment analysis: *ptprdb* (hypertrophic, +18.5%), *ptenb* (hyperplastic, -7.9%), and *srpx* (hyperplastic, -0.9%) (Table 1). These represent suggestive hits that may warrant further investigation in stable mutant lines. Together, these findings demonstrate image-based profiling is an effective and scalable approach for identifying regulators of adipose morphology.

A hypertrophic growth bias of subcutaneous adipose in *foxp1b* and double *foxp1a;foxp1b* zebrafish mutants

Notably, targeting *foxp1b* (one of two zebrafish *foxp1* genes) induced hypertrophic SAT morphology (Fig. 5). Foxp1 is required for maintaining undifferentiated and self-renewing stem and progenitor cells in multiple tissue compartments (40–44). Further, in mouse, Foxp1 inhibits brown adipogenesis and browning of white adipocytes (45). However, roles for Foxp1 in establishing hyperplastic/hypertrophic morphology of adipose is unknown. To further study the role of Foxp1, we generated new loss-of-function alleles for both *foxp1* genes (*foxp1a* and *foxp1b*) in zebrafish (Fig. 6A). We targeted the highly conserved Forkhead domain (FHD) (Fig. 6B), which is essential for DNA binding and therefore Foxp1 transcriptional activity (46,47) (Fig. 6B).

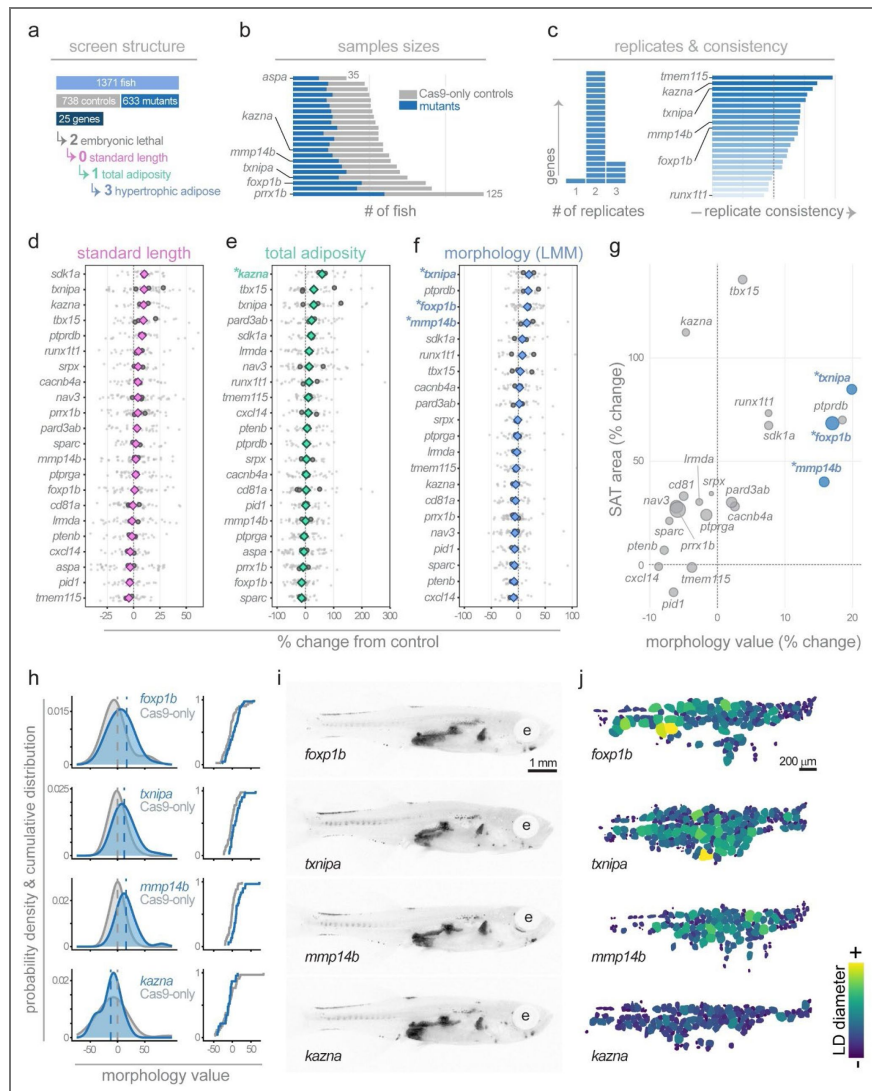


Figure 5. A targeted CRISPR screen in zebrafish identifies genes regulating hypertrophic or hyperplastic adipose morphology.

A. Overview of CRISPR screen structure. In total, 1,371 fish were screened (738 Cas9-only controls and 633 mutants) across 25 candidate genes. Two genes (*phb* and *rera*) were lethal before 5 dpf. Of the remaining genes, zero altered standard length, one (*kazna*) significantly increased total adiposity, and three (*txnipa*, *mmp14b* and *foxp1b*) produced significant hypertrophic subcutaneous adipose morphology. **B.** Sample sizes per gene. Grey bars indicate Cas9-only controls; blue bars indicate mutants. *aspa* had the smallest and *prx1b* the largest sample size. **C.** Left panel: number of experimental replicates per gene. The majority of targets had two replicates; *aspa* had one replicate, and *nav3*, *prx1b*, *tbx15* and *txnipa* had three. Right panel: replicate consistency scores, representing agreement in effect direction and magnitude across replicates. Genes are ordered by consistency score. **D-F.** Forest plots showing effect sizes (% change from Cas9-only controls) for standard length (d, magenta), total adiposity (e, green) and morphology value (f, blue). Small grey dots represent individual mutant fish; medium grey circles represent experiment means; coloured diamonds represent gene-level estimates from linear mixed models (LMM) with experiment as a random effect. Morphology value represents the deviation in lipid droplet Feret diameter from that expected given total depot area, with positive values indicating hypertrophy and negative values indicating hyperplasia. Asterisks denote BH-adjusted $p < 0.05$. **G.** Scatter plot of depot (SAT) lipid area (% change from controls) versus morphology value (LMM estimate, % change from controls) for each gene. Point size reflects sample size. Quadrants indicate the combination of depot size and cell morphology phenotype (e.g., bigger depot with hypertrophic cells). **H.** Probability density functions (left) and cumulative distribution functions (right) of morphology values for *foxp1b*, *txnipa*, *mmp14b* and *kazna* F0 mutants (blue) compared with matched Cas9-only controls (grey). Dashed lines indicate group means. **I.** Representative Nile Red images of *foxp1b*, *txnipa*, *mmp14b* and *kazna* F0 mutants showing total adiposity. e, eye. Scale bar: 1 mm. **J.** Segmented subcutaneous adipose lipid droplets from representative mutant fish, colour-coded by lipid droplet (LD) diameter. Scale bar: 200 μ m.

We generated two alleles – *foxp1a*^{ed116} and *foxp1b*^{ed125} – which each contained a frameshift and premature stop codon within the FHD (Fig. 6B). qRT-PCR revealed reduced *foxp1a* or *foxp1b* mRNA in the respective single mutants, with no apparent mRNA upregulation of the paralogous *foxp1* gene, or other members of the Foxp family (Suppl. Fig. 4). Western blots revealed a significant reduction of Foxp1 protein in each of the single mutants, and in *foxp1a;foxp1b* double mutants suggesting severe loss-of-function alleles (Fig. 6C & Suppl. Fig. 4). At approximately one-month of age (~10 mm SL), *foxp1b*^{ed125} and double *foxp1a;foxp1b* mutants were significantly smaller than either wild-type control or *foxp1a*^{ed116} fish (Fig. 6E). Analysis of baseline lipid storage using Nile Red revealed that *foxp1a;foxp1b* double mutants had significantly lower normalised adipose area than controls, while *foxp1b* single mutants showed a non-significant trend (Fig. 6D,F). Critically, despite their smaller body size, both *foxp1b*^{ed125} and *foxp1a;foxp1b* mutants had significantly larger LDs characteristic of hypertrophic morphology (Fig. 6G). This dissociation between body size (smaller) and LD size (larger) argues against a simple developmental delay: if these mutants were merely developmentally delayed, we would expect smaller fish to have smaller LDs, as observed during normal development (Fig. 3G). Instead, *foxp1b* mutants seem to display fundamentally altered adipose growth properties. Analysis of blood metabolites revealed *foxp1a;foxp1b* double mutants were hyperglycaemic (Suppl. Fig. 5), and strikingly, in *foxp1a;foxp1b* double mutants there was increased lipid accumulation in the liver (Fig. 6D & Suppl. Fig. 5D). At adulthood, fish size and lipid storage within adipose in *foxp1a;foxp1b* double mutants had largely caught up with wild-type control fish (adiposity at ~80% of wild-type levels) (Suppl. Fig. 6). These findings show that stable *foxp1b* mutants have hypertrophic adipose along with metabolic dysfunction.

Spatial analysis reveals profoundly blunted hypertrophic responses to high-fat diet in *foxp1b* mutants

Our previous analyses showed that *foxp1b* mutants display a bias toward hypertrophic growth during developmental expansion of SAT. To determine whether this bias impacts the adipose response to dietary challenge, we subjected *foxp1a* and *foxp1b* mutants to a two-week high-fat diet (HFD), a condition previously shown to induce SAT expansion (16). All *foxp1* mutant genotypes exhibited reduced overall HFD-induced adipose expansion compared to control fish, indicating a general impairment in expansion capacity (Suppl. Fig. 7). To characterise the HFD response, we performed spatial analysis by binning the anterior-posterior (AP) axis into 200 µm strata, enabling comparison of LD morphology at anatomically equivalent positions (Fig. 6H-M). Using paired analysis (same fish imaged on control diet at 35 dpf and after 14 days HFD at 49 dpf) with linear mixed-effects models, we quantified diet effects across all strata. First, we confirmed that on a control diet, *foxp1b* mutants had significantly larger LDs than wild-type across strata 1–5 (ranging from +22.2 µm in stratum 1 to +17.8 µm in stratum 5; all FDR-adjusted $p < 0.05$; Fig. 6I). By contrast, *foxp1a* mutants showed no baseline hypertrophy at any stratum (all $p > 0.10$; Suppl. Figs. 8 & 9). Second, spatial analysis revealed that *foxp1b* mutants show profoundly blunted hypertrophic responses to HFD. In wild-type fish, HFD induced robust LD hypertrophy across all 15 strata (range: +17.7 to +28.1 µm; all FDR-adjusted $p < 0.025$; Fig. 6J). However, *foxp1b* mutants showed a strikingly different pattern: in anterior/oldest strata (1–5), mutants showed significant but attenuated HFD responses (+7.7 to +12.2 µm; ~57% weaker than wild-type; FDR-adjusted $p < 0.05$ for strata 3–5), while in posterior/newer strata (7–15), there was essentially no hypertrophic response (effect sizes declining from +6.8 µm to +0.4 µm; all FDR-adjusted $p > 0.068$; Fig. 6J). Overall, averaging across all strata, wild-type LDs increased by +24.4 µm in response to HFD ($p < 0.0001$), whereas *foxp1b* mutant LDs increased by only +7.7 µm ($p = 0.020$) – representing a 68% reduction in hypertrophic capacity (Fig. 6K). *foxp1a* mutants had no baseline hypertrophy on control diet (all strata $p > 0.10$; Fig. 6L), but still exhibited significantly blunted HFD responses (~35% reduction compared to wild-type; $p = 0.023$; Fig. 6M), demonstrating that an impaired adaptive response can occur independently of baseline LD size. Interestingly, *foxp1a* mutants also lost the spatial gradient of HFD responsiveness observed in wild-type fish (uniform +14.4 µm across all strata vs wild-type range of +16.6 to +26.4 µm anteriorly to posteriorly), suggesting Foxp1a may regulate spatial coordination within the tissue. Together, these spatial analyses reveal

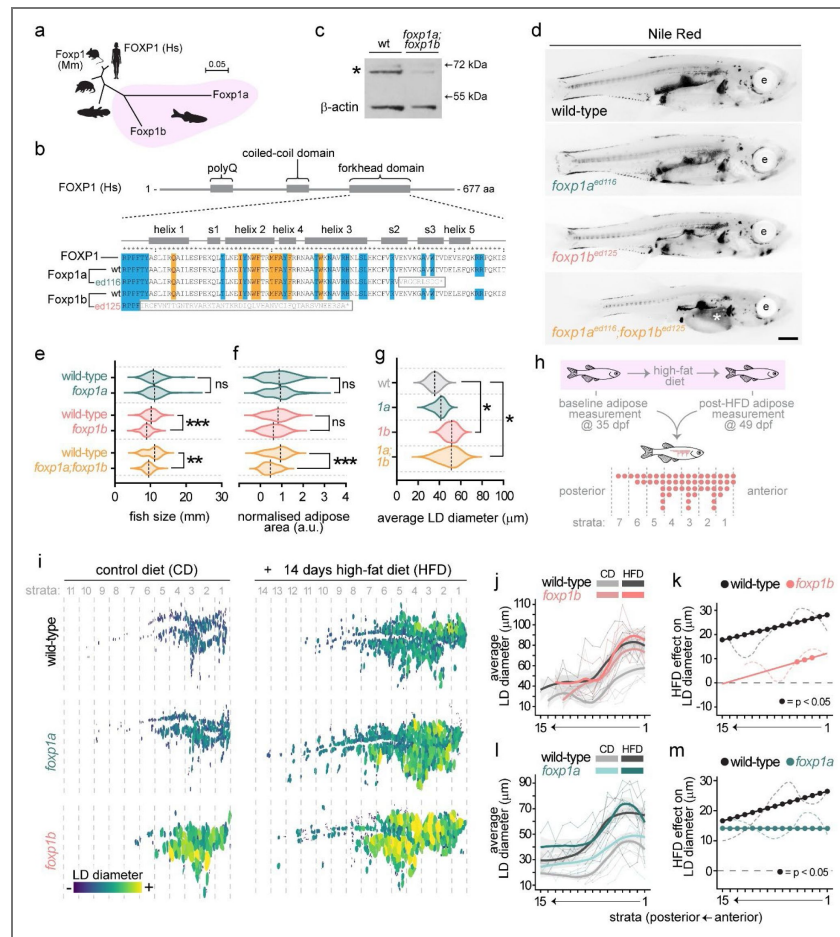


Figure 6. Stable *foxp1b* zebrafish mutants have hypertrophic adipose but undergo severely reduced hypertrophic remodelling in response to a high-fat diet.

A. Phylogenetic tree showing relatedness of zebrafish Foxp1a and Foxp1b amino acid sequences to human, mouse, opossum and coelacanth Foxp1. Scale bar indicates substitutions per site. **B.** Overview of human FOXP1 domain structure showing polyglutamine (polyQ), coiled-coil and forkhead domains. Zoomed view of the DNA-binding forkhead domain showing structural features including helices (helix 1–5) and beta-strands (s1–s3). Amino acids involved in DNA binding are highlighted in blue; residues at the FOXF domain-swapped dimer interface are highlighted in orange. Wild-type zebrafish Foxp1a and Foxp1b sequences are aligned to human FOXP1, along with the ed116 (*foxp1a*) and ed125 (*foxp1b*) mutant alleles. Grey boxes indicate the addition of nonsense peptide sequence followed by a premature stop codon. **C.** Western blot showing reduction of Foxp1 protein in *foxp1a*; *foxp1b* double mutants compared to wild-type. β -actin serves as a loading control. Asterisk indicates the Foxp1 band. **D.** Nile Red fluorescence images showing adipose lipid distribution (black signal) in wild-type, *foxp1a*^{ed116}, *foxp1b*^{ed125} and double *foxp1a*^{ed116}; *foxp1b*^{ed125} zebrafish mutants. e, eye. Asterisk indicates lipid accumulation within the liver in double mutants. Scale bar is 1 mm. **E.** Violin plots of fish size (standard length, mm) in *foxp1a*^{ed116}, *foxp1b*^{ed125} and double *foxp1a*^{ed116}; *foxp1b*^{ed125} mutants compared to wild-type siblings. **F.** Violin plots of normalised adipose area in the same genotypes. **G.** Violin plots of average lipid droplet (LD) diameter in the same genotypes. **H.** Schematic of the high-fat diet (HFD) feeding experiment. Zebrafish were Nile Red-imaged at 35 dpf to establish baseline adipose measurements, then subjected to a 14-day HFD (2-hour daily immersion in 5% chicken egg yolk) or control diet (2-hour daily immersion in system water), in addition to normal feeding. Post-diet Nile Red imaging was performed at 49 dpf. Subcutaneous adipose tissue was divided into anterior–posterior strata for spatial analysis. 200 μ m strata are numbered anterior (1) to posterior. **I.** Segmented subcutaneous adipose LDs from representative wild-type, *foxp1a* and *foxp1b* fish on control diet (left) and after HFD (right), colour-coded by LD diameter. Strata boundaries (200 μ m) are indicated by dashed lines. **J.** Average LD diameter per stratum for wild-type and *foxp1b* fish on control diet (grey or salmon) and HFD (black or pink). Thin lines represent individual fish; thick lines represent group means. **K.** HFD effect on LD diameter per stratum (HFD minus control diet) for wild-type (black) and *foxp1b* (pink). Filled circles indicate strata where the HFD effect is significant (BH adjusted $p < 0.05$). **L.** As in (J), comparing wild-type and *foxp1a*. **M.** As in (K), comparing wild-type (black) and *foxp1a* (teal). Statistical tests in (E–G) were one-way ANOVA followed by Tukey’s HSD post-hoc test. ** $p < 0.01$, *** $p < 0.001$, ns = not significant.

that *foxp1b* mutants have intrinsically impaired capacity for diet-induced hypertrophic remodelling, while *foxp1a* mutants show a distinct phenotype affecting the magnitude and spatial patterning of adaptive responses. These findings demonstrate that Foxp1 paralogs play non-redundant roles in regulating both constitutive adipose morphology and adaptive responses to nutritional challenge.

Discussion

In this study, we (i) identified candidate human genes implicated in hypertrophic and hyperplastic adipose morphology, (ii) established an image-based profiling method for assessing adipose morphology in zebrafish, (iii) conducted an in vivo CRISPR screen to functionally evaluate the roles of 25 candidate genes with rigorous statistical validation, (iv) identified three genes that induce hypertrophic adipose morphology, (v) generated stable *foxp1a* and *foxp1b* zebrafish mutant lines that recapitulate phenotypes observed in the F0 CRISPR experiments and revealed paralogue-specific roles in adipose biology, and (vi) discovered through spatial analysis that *foxp1b* mutants have intrinsically impaired capacity for diet-induced hypertrophic remodelling. Together, this work demonstrates a scalable, rigorous screening platform to identify regulators of adipose morphology with clear translational potential.

Candidate gene identification from human data

We employed a multi-step process to identify candidate morphology genes from human transcriptomic data. As GWAS for adipose morphology has only been conducted in relatively small cohorts (48,49), we instead used published data identifying DEGs enriched in SAT characterised by large or small adipocytes (27). Clustering of the morphology DEG expression correlations with cardiometabolic traits separated genes into ‘healthy’ and ‘unhealthy’ profiles (Suppl. Fig. 2 4); genes associated with larger adipocytes generally aligned with adverse metabolic traits such as increased BMI and WHR in accordance with adipocyte size increasing with BMI (50). However, this pattern was not absolute—some candidates diverged from this axis (Suppl. Fig. 2 4), representing potentially interesting regulators of adipose morphology that are uncoupled from traditional metabolic or disease states. We focussed on candidate genes highly expressed in adipocyte stem and progenitor cells (ASPCs), reasoning that progenitor function influences both hyperplastic potential and overall remodelling capacity. However, we note that adipose remodelling is likely determined by multiple cell types beyond progenitors, including the adipose microenvironment (51,52).

An image-based platform for profiling adipose morphology

We established a rapid, scalable, stereomicroscope-based imaging pipeline for profiling zebrafish adipose morphology. Our imaging resolution was sufficient to detect large LDs but not smaller secondary LDs found in zebrafish adipocytes (52), a phenomenon also observed in mouse adipocytes (53). Nevertheless, our methodology reliably captured primary LDs and allowed us to use LD size as a proxy for overall adipocyte size. We adapted the morphology value framework developed by the Arner lab for human adipose (35), and observed strikingly similar relationships in zebrafish SAT, including a curvilinear relationship between LD size and depot size and an inverse correlation between LD number and LD size. These parallels suggest that the fundamental growth dynamics of zebrafish adipose are conserved with mammals, supporting zebrafish as a relevant model for studying adipose biology.

Statistical validation and batch effect control

A key advance of our platform is the implementation of rigorous statistical validation to control for batch effects - a common source of false positives in genetic screens. We employed two complementary approaches: Kolmogorov–Smirnov (KS) tests to compare morphology value distributions, and linear mixed models (LMMs) to assess LD size while accounting for experimental batch as a random effect. Critically, we performed both pooled KS tests (combining data across experiments) and stratified KS tests (testing within each experiment separately, then

combining p-values using Fisher's method). This stratified approach validates that phenotypic effects are reproducible within independent experiments rather than driven by between-batch variability. Genes were considered robust hits only if significant in both stratified KS and LMM analyses after FDR correction—a conservative threshold that substantially reduces false discovery risk. Using these criteria, we identified three genes (*txnipa*, *mmp14b*, *foxp1b*) as robust hypertrophic hits. Three additional genes (*ptprdb*, *ptenb*, *srpx*) showed suggestive effects in pooled analyses but were not validated by stratified tests, indicating potential batch artifacts or weaker effects requiring larger sample sizes. Two genes (*tbx15*, *tnfrsf115*) showed significant distributional shifts in stratified KS without corresponding LMM intercept effects; notably, *tbx15* displayed a significant slope effect, indicating size-dependent rather than uniform phenotypes. This statistical framework - combining distribution-sensitive tests with mixed models and stratified validation - provides a template for future high-throughput screens where batch effects are unavoidable.

Functional insights from screen hits

Txnipa (*txnipa*) F0 CRISPR mutants displayed hypertrophic morphology. TXNIP (Thioredoxin-Interacting Protein) is a well-characterised regulator of oxidative stress, metabolism, and inflammation that binds and inhibits thioredoxin, leading to increased production of reactive oxygen species and oxidative stress (54). TXNIP links nutrient sensing to inflammatory signalling by responding to glucose and regulating NLRP3 inflammasome activation (54). The ROS/TXNIP/NLRP3 pathway impairs insulin signalling, a major driver of adipogenesis and hyperplastic adipose growth (55), potentially explaining why *txnipa* loss biases toward hypertrophy. Additionally, vitamin D increases TXNIP expression (56), and plays a major role in energy storage in zebrafish (57), suggesting nutrient-sensing pathways may coordinate morphological outcomes.

Mmp14b F0 zebrafish mutants also exhibited hypertrophic adipose. MMP14 (Matrix Metalloproteinase 14) is a pericellular collagenase and plays a key role in ECM remodelling. In mice, MMP14 deficiency stiffens the adipose matrix and restricts expansion, resulting in smaller adipocytes (58), while overexpression of *Mmp14* increased the size of adipocytes (59). Our zebrafish findings – hypertrophy upon *mmp14b* loss – appear to contrast with these mouse phenotypes and warrant further investigation. Notably, zebrafish possess two *mmp14* paralogues, therefore functional redundancy or divergence between paralogues may underlie species-specific differences. Future studies targeting both paralogues simultaneously would clarify this.

Paralogue selection: strengths and limitations

Our screen targeted single paralogues based on DIOPT orthology scores, selecting the zebrafish gene with highest predicted orthology to each human candidate. This approach has important limitations: orthology scores predict sequence conservation but not necessarily functional equivalence, meaning the “most orthologous” paralogue may not carry the relevant adipose function. Indeed, our three candidate hit genes have multiple zebrafish paralogues (FOXP1, TXNIP, MMP14), and testing only one paralogue means we may have missed genuine hits where the alternative paralogue is functionally relevant. Our *Foxp1* findings both validate this approach and reveal its limitations. The higher-scoring paralogue *foxp1b* (DIOPT score 13/19) showed the more severe baseline phenotype, consistent with our prioritisation. However, *foxp1a* (DIOPT score 5/19), tested subsequently, revealed a distinct and biologically significant phenotype affecting spatial patterning of HFD responses - a finding that would have been missed had we not pursued secondary validation. For future screens where comprehensive hit identification is the goal, multiplexed F0 targeting of all paralogues may be valuable (67), though this approach may complicate interpretation of paralogue-specific phenotypes.

Non-redundant roles for Foxp1 paralogues in adipose biology

Foxp1 is known to maintain stem and progenitor populations across multiple tissues including the hair follicle, haematopoietic niche, mammary gland, and cerebral cortex (40,41,43,44). We hypothesised that Foxp1 may be required in adipose progenitors for hyperplastic growth, with its loss resulting in compensatory hypertrophy. *Foxp1b* mutants displayed constitutive hypertrophy at baseline - smaller fish with paradoxically larger LDs - along with a reduced hypertrophic response to HFD together with reduced adipose coverage, hyperglycaemia, and hepatic steatosis in double mutants. In contrast, *foxp1a* mutants showed no baseline morphological defects but also exhibited impaired adaptive responses to HFD and a breakdown in anterior-posterior patterning in the adipose tissue. These data suggest both *foxp1* paralogues regulate adaptive responses to HFD, but also regulate distinct aspects of adipose biology: Foxp1b controlling constitutive morphology and Foxp1a regulating spatial patterning.

Impaired adaptive capacity rather than developmental ceiling

A key finding from our spatial analysis is that *foxp1b* mutants have intrinsically impaired capacity for diet-induced hypertrophy, rather than having reached a developmental size limit. Several observations support this conclusion. First, the blunted HFD response in *foxp1b* mutants extended throughout the entire anterior-posterior axis (68% overall reduction), including posterior regions containing smaller, newer LDs. If anterior adipocytes had simply reached maximum size, we would expect normal responses posteriorly - but instead we observed complete loss of response in posterior strata. Second, and most strikingly, wild-type LDs on HFD actually surpassed *foxp1b* mutant sizes in posterior strata, demonstrating that these sizes are not developmentally limiting. Third, *foxp1a* mutants showed blunted HFD responses (~35% reduction) despite having no baseline hypertrophy, proving that impaired adaptive capacity can occur independently of baseline LD size. These findings have important implications. They suggest that both *foxp1a* and *foxp1b* mutants may fail to perceive or respond to HFD-induced molecular cues, potentially implicating Foxp1 in metabolic sensing pathways (66). The distinct *foxp1a* phenotype - loss of spatial gradients in HFD responsiveness - suggests this paralogue may coordinate regional heterogeneity within adipose tissue. Testing whether other hypertrophic mutants identified in our screen (*txnipa*, *mmp14b*) also show impaired adaptive responses would determine whether this reflects a general principle linking developmental morphology to later plasticity.

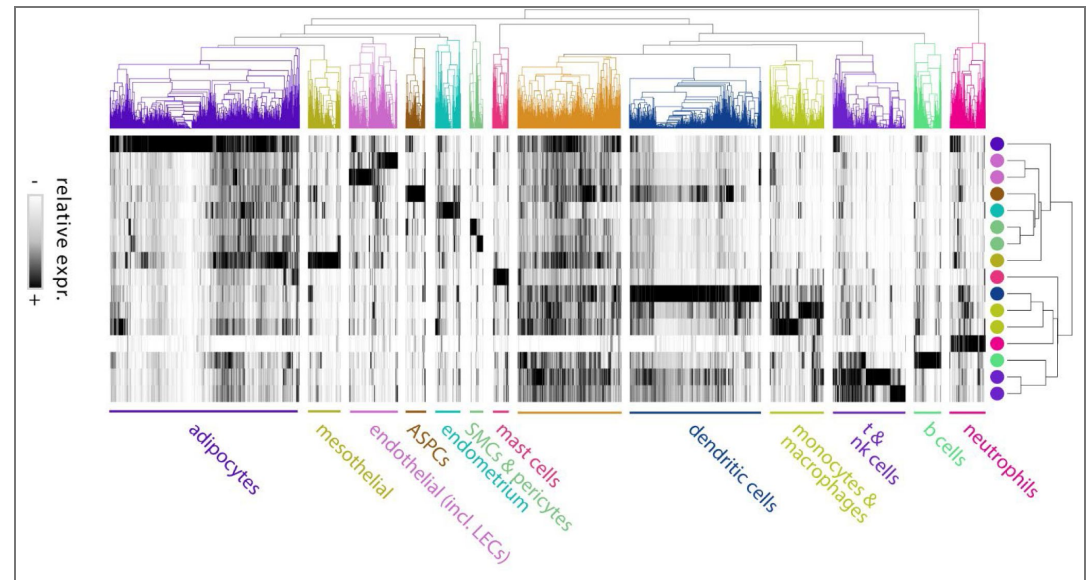
Comparison with mammalian Foxp1 studies

In mouse, adipose-specific Foxp1 deletion increases browning of white adipose tissue and enhances thermogenesis, protecting against diet-induced obesity (45). Our zebrafish findings - where whole-organism *foxp1b* loss causes metabolic dysfunction rather than protection - appear contradictory but likely reflect important experimental differences. First, Liu et al. studied browning-prone inguinal WAT, whereas zebrafish adipose, as in an ectotherm, is not traditionally considered thermogenic - though recent evidence shows zebrafish epicardial adipose can be thermogenic (15). In browning-resistant depots, more commonly observed in humans, Foxp1 loss might produce phenotypes more similar to our zebrafish findings. Second, our whole-organism mutants also lack hepatic Foxp1 which regulates gluconeogenesis (65); the metabolic dysfunction we observe may reflect hepatic rather than adipose-autonomous effects. Tissue-specific approaches in zebrafish would help distinguish these contributions.

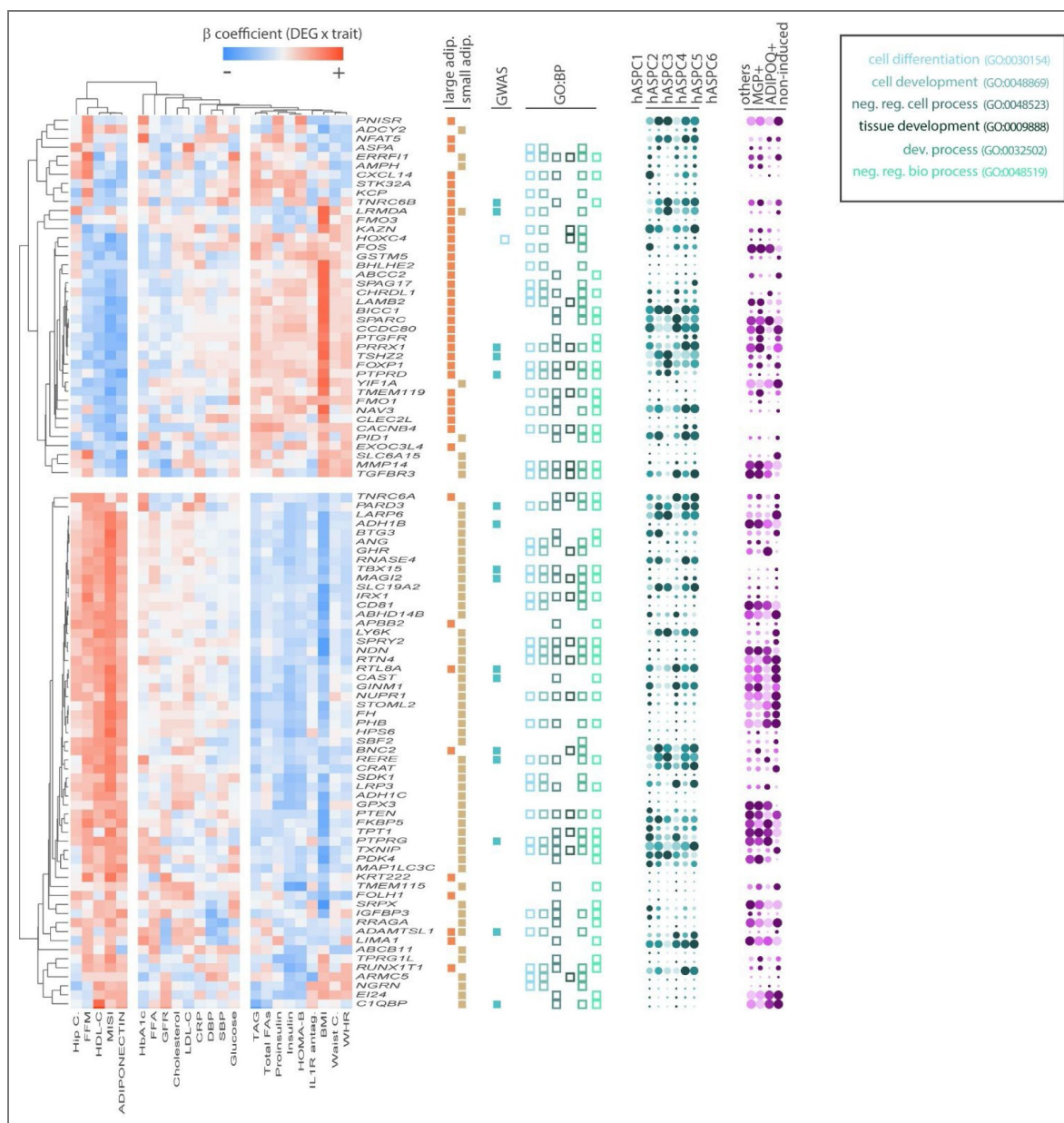
Platform scalability and translational potential

Our screening platform offers several advantages for systematic functional genomics. First, it is highly scalable: we screened 1,371 fish across 25 genes with sufficient statistical power (>80% for large effects) in each case. The imaging pipeline requires only standard stereomicroscopy, and analysis is fully automated. Second, it is modular: the same fish can be assessed for multiple phenotypes (body size, total adiposity, morphology, spatial patterning), and the platform readily accommodates dietary or pharmacological challenges. Third, results translate meaningfully to mammalian biology, as demonstrated by the conservation of morphology value relationships.

Compared to mammalian screening platforms, zebrafish offer dramatically reduced costs and timelines while enabling whole-organism assessment of gene function. However, limitations include the lack of tissue-specific genetic tools comparable to Cre-lox systems, potential paralogue complications, and the possibility that some mammalian adipose biology (particularly thermogenesis) may not be fully conserved. Nevertheless, our identification of genes with established human disease relevance (*TXNIP*, *MMP14*, *FOXP1*) demonstrates clear translational line-of-sight from zebrafish discoveries to human adipose biology.

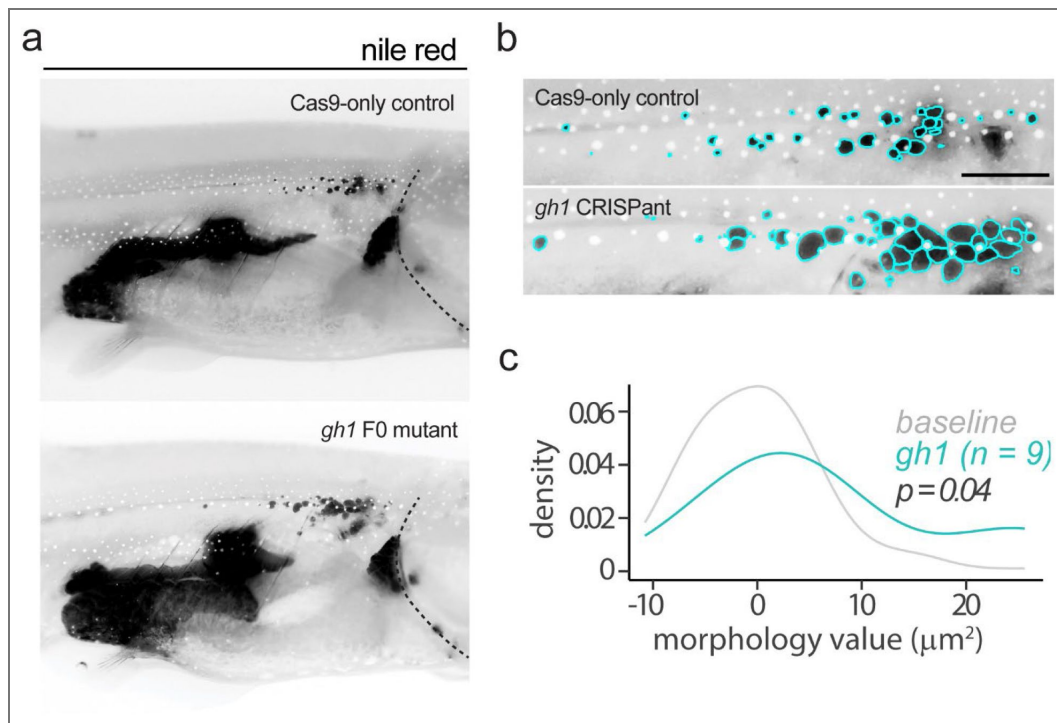


Supplemental Figure 1. Expression clustering of candidate adipose morphology genes across human white adipose tissue cell types. Heatmap showing relative expression (log-normalised counts) of 2,980 candidate morphology-associated genes across 16 human WAT cell types from Emont et al., 2022. Genes without annotations or detectable expression in the single-cell dataset were excluded. Hierarchical clustering of genes (rows) and cell types (columns) was performed using average linkage with Pearson correlation distance. Cell types are colour-coded and labelled below: adipocytes, mesothelial cells, endothelial cells (including lymphatic endothelial cells, LECs), adipose stem and progenitor cells (ASPCs), smooth muscle cells (SMCs) and pericytes, endometrium, mast cells, dendritic cells, monocytes and macrophages, T and natural killer (NK) cells, B cells, and neutrophils. Coloured dots (right) correspond to cell type identity in the dendrogram (top).



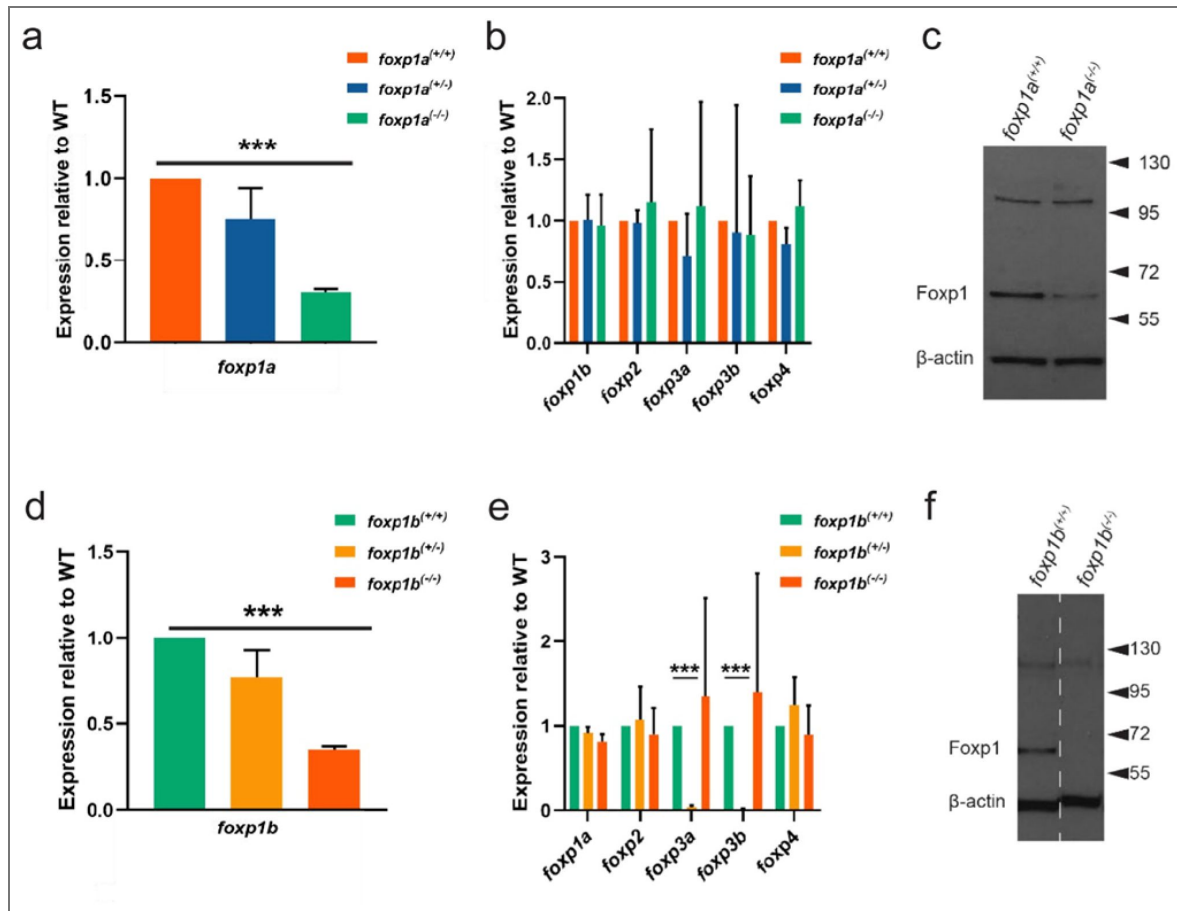
Supplemental Figure 2. Integrative analysis of adipose morphology genes across cardiometabolic traits, GWAS loci, Gene Ontology terms, and single-cell expression profiles.

The 102 candidate morphology genes enriched in adipose stem and progenitor cells (ASPCs) were characterised across multiple data sources. Left panel: Heatmap of β coefficients from linear regression of gene expression in human subcutaneous adipose tissue (SAT) against 23 cardiometabolic traits from the METSIM cohort (Civelek et al., 2017). Blue indicates negative and red indicates positive associations. Genes and traits are hierarchically clustered. Traits include measures of body composition and adiposity (BMI, Waist C, WHR, Hip C, FFM), lipids (HDL-C, LDL-C, cholesterol, TAG, total FAs, FFA), glycaemia and insulin sensitivity (HbA1c, glucose, insulin, proinsulin, HOMA-B, MISI), inflammation (CRP, IL-1R antagon.), blood pressure (SBP, DBP), renal function (GFR) and adipokines (adiponectin). Centre columns (left to right): Adipocyte size association — brown and orange bars indicate genes enriched in large or small adipocytes, respectively (Honecker et al., 2022). GWAS overlap - teal boxes indicate genes at genome-wide significant loci for BMI or WHR-adjusted BMI. GO biological process - colour-coded boxes indicate membership in enriched GO terms as shown in the legend: cell differentiation (GO:0030154), cell development (GO:0048869), negative regulation of cellular process (GO:0048523), tissue development (GO:0009888), developmental process (GO:0032502) and negative regulation of biological process (GO:0048519). Right panel: Dot plots showing expression of each gene across human WAT single-cell clusters (Emont et al., 2022 & Yang Loureiro et al., 2023). Dot size represents percentage of cells expressing; colour intensity represents scaled mean expression. Cell populations shown are: human ASPC subclusters (hASPCs1-6), MGP+, ADIPOQ+ non-induced progenitors and others.



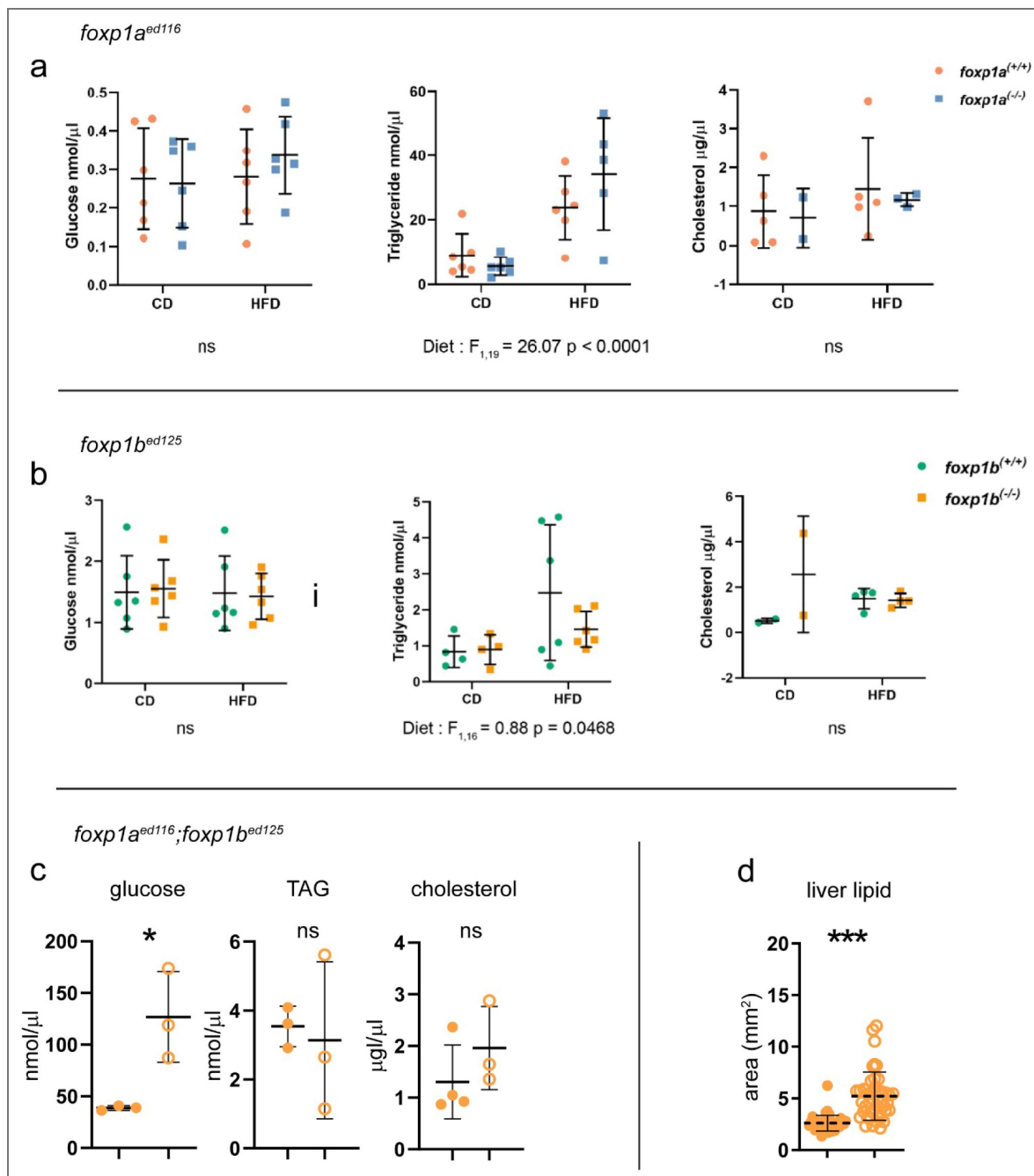
Supplemental Figure 3. Image-based adipose morphology profiling detects hypertrophic lipid droplet expansion in *gh1* CRISPR mutants.

A. Nile Red fluorescence images showing adiposity in a representative Cas9-only control (top) and *gh1* F0 CRISPR mutant (bottom) at ~1 month of age. Black signal denotes Nile Red-positive neutral lipid. Dotted line indicates operculum. **B.** High-magnification images of subcutaneous adipose tissue (SAT) from Cas9-only control and *gh1* CRISPR zebrafish. Cyan outlines show automated segmentation of individual lipid droplets (LDs). **C.** Kernel density plot of morphology values for *gh1* mutants ($n = 9$, cyan) compared to baseline wild-type fish ($n = 194$, grey). Morphology values were significantly increased in *gh1* mutants (Kolmogorov-Smirnov test, $p = 0.04$), indicating a shift toward hypertrophic morphology.



Supplemental Figure 4. Stable *foxp1a* and *foxp1b* mutant alleles lead to reduced transcript and protein expression.

A. qRT-PCR showing significantly reduced *foxp1a* mRNA expression in homozygous *foxp1a*^{ed116} mutants compared to heterozygotes and wild-type siblings. **B.** Expression of other *foxp1* family genes in *foxp1a*^{ed116} mutants shows no evidence of compensatory upregulation. **C.** Western blot of Foxp1 protein in *foxp1a* wild-type (+/+) and *foxp1a*^{ed116} mutant caudal fin lysates. Foxp1 protein is markedly reduced in mutants. β-actin serves as a loading control. **D.** qRT-PCR showing significantly reduced *foxp1b* mRNA in *foxp1b*^{ed125} mutants compared to heterozygotes and wild-type siblings. **E.** Expression of other *foxp* family genes (*foxp1a*, *foxp2*, *foxp3a*, *foxp3b*, *foxp4*) in *foxp1b*^{ed125} mutants. *foxp3a* and *foxp3b* are significantly downregulated in heterozygotes, however this likely reflects defective amplification in those samples as homozygous mutants show no change. **F.** Western blot of Foxp1 protein in *foxp1b* wild-type (+/+) and *foxp1b*^{ed125} caudal fin lysates, showing reduced Foxp1 protein in mutants. β-actin was used as a loading control. Error bars represent standard error of the mean (SEM). ***p < 0.001, by unpaired two-tailed t-test.

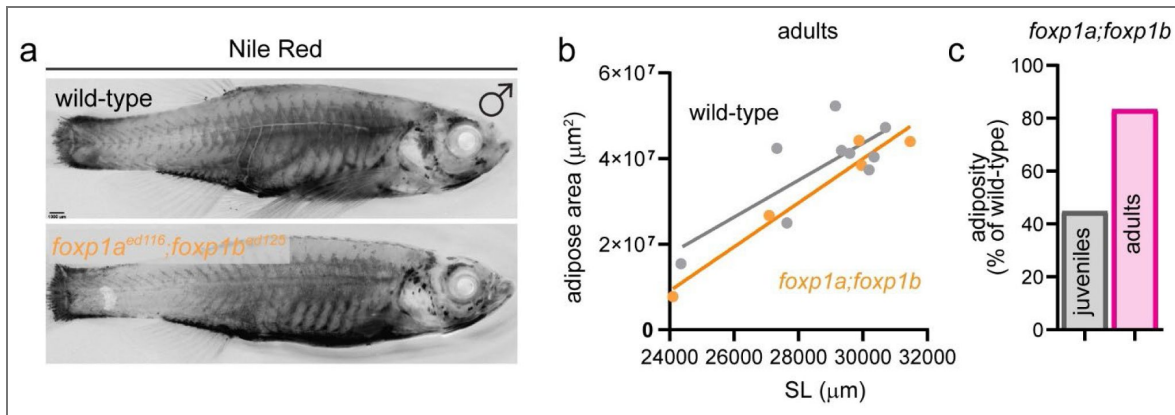


Supplemental Figure 5. Metabolic phenotypes of *foxp1a*, *foxp1b*, and *foxp1a;foxp1b* mutant zebrafish following dietary challenge.

A. Plasma glucose, triglyceride (TAG), and cholesterol levels in *foxp1a^{ed116}* mutants and wild-type siblings under control diet (CD) or high-fat diet (HFD) conditions. HFD induced a significant increase in TAG across genotypes (two-way ANOVA: $F_{1,19} = 26.07$, $p < 0.0001$). No significant genotype or diet effects were observed for glucose or cholesterol. **B.** Plasma glucose, TAG, and cholesterol in *foxp1b^{ed125}* mutants and wild-type controls under CD and HFD. TAG levels were modestly influenced by diet (two-way ANOVA: $F_{1,16} = 0.88$, $p = 0.0468$), with no significant effects detected for glucose or cholesterol. **C.** Circulating glucose, TAG, and cholesterol levels in *foxp1a^{ed116};foxp1b^{ed125}* double mutants. Double mutants exhibited significantly elevated glucose levels ($p < 0.05$, unpaired t-test), while TAG and cholesterol levels were unchanged. **D.** Quantification of liver lipid accumulation in *foxp1a^{ed116};foxp1b^{ed125}* double mutants revealed significantly increased hepatic lipid deposition compared to controls ($p < 0.001$, unpaired t-test). Error bars represent standard deviation (SD). ns = not significant.

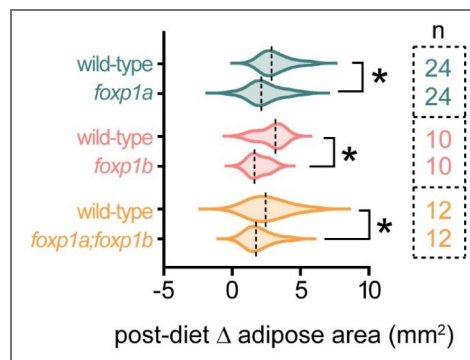
Supplemental Figure 6. Adiposity in adult *foxp1a*^{ed116};*foxp1b*^{ed125} zebrafish mutants.

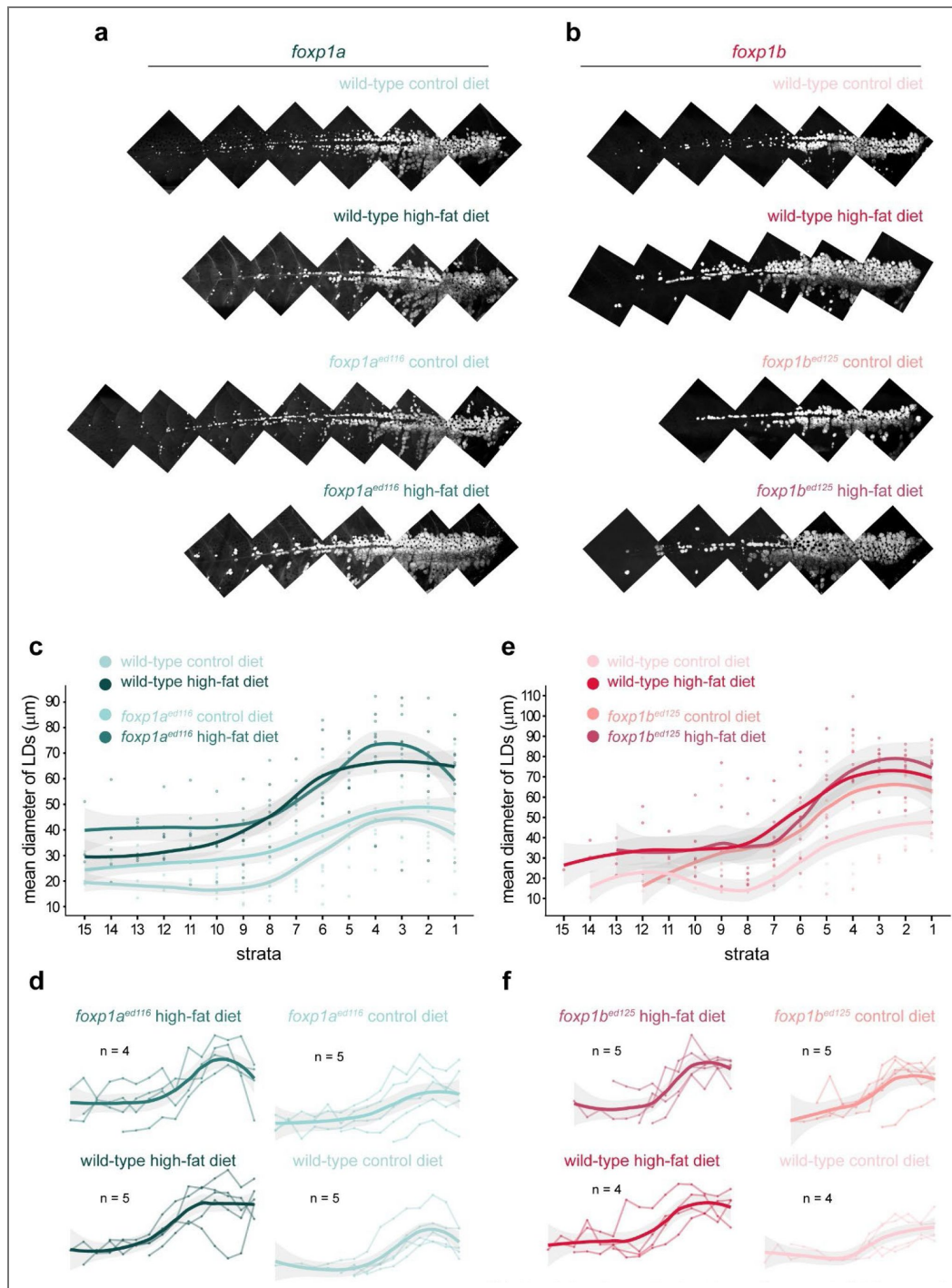
A. Representative Nile Red-stained images of adult male wild-type and *foxp1a*^{ed116};*foxp1b*^{ed125} double mutant zebrafish. Lipid-rich adipose tissue appears as dark black signal. Scale bar = 1 mm. **B.** Total adipose area plotted against standard length (SL, μm) in adult wild-type (grey) and *foxp1a*;*foxp1b* mutants (orange). A regression line is fitted for each group, showing reduced adiposity in mutants relative to controls (μm^2). **C.** Comparison of adiposity in *foxp1a*;*foxp1b* mutants relative to wild-type siblings at juvenile and adult stages. While adiposity was markedly reduced in juveniles, it recovered to ~80% of wild-type levels by adulthood.



Supplemental Figure 7. Impaired adipose expansion in all *foxp1* mutant genotypes following high-fat diet.

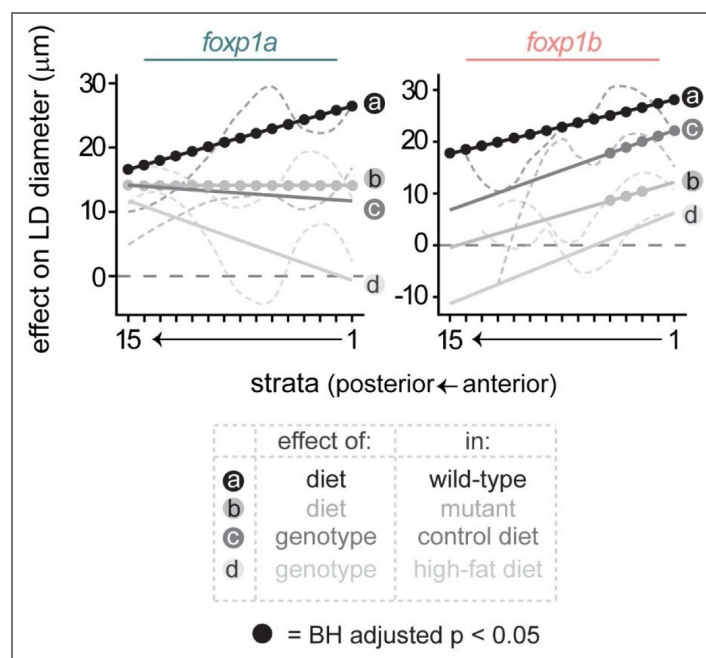
Violin plots showing change in subcutaneous adipose area (Δ adipose area) after two weeks of high-fat diet (HFD) exposure in wild-type and *foxp1a*^{ed116}, *foxp1b*^{ed125}, and double mutant *foxp1a*^{ed116};*foxp1b*^{ed125} zebrafish. All three mutant genotypes exhibited significantly reduced adipose expansion compared to wild-type controls, indicating impaired capacity to remodel adipose in response to nutritional excess. Sample sizes are shown at right. Asterisks indicate statistically significant differences (* $p < 0.05$), by one-way ANOVA followed by Tukey's HSD post-hoc test.





Supplemental Figure 8. Spatial analysis of lipid droplet size along the anterior-posterior axis in *foxp1a* and *foxp1b* mutants following high-fat diet challenge.

A. Representative Nile Red fluorescence montages of the subcutaneous adipose depot in wild-type and *foxp1a^{ed116}* fish under control diet (CD) and high-fat diet (HFD) conditions. **B.** Representative Nile Red montages of wild-type and *foxp1b^{ed125}* fish under CD and HFD conditions. **C.** Mean lipid droplet (LD) diameter (μm) per 200 μm stratum along the anterior-posterior axis for wild-type (grey) and *foxp1a^{ed116}* mutants (teal) under CD (light) and HFD (dark). Lines represent LOESS fits with shaded 95% confidence intervals; dots represent individual stratum means per fish. Strata are numbered from anterior (1) to posterior (15). **D.** Individual fish traces for each group in the *foxp1a* experiment. Thin lines represent individual fish; thick lines with shading represent group means. Sample sizes are indicated. **E.** Mean LD diameter per stratum for wild-type (grey) and *foxp1b^{ed125}* mutants (pink/red) under CD and HFD, plotted as in (C). **F.** Individual fish traces for each group in the *foxp1b* experiment, plotted as in (D).



Supplemental Figure 9. Decomposition of diet and genotype effects on lipid droplet size across the anterior-posterior axis in *foxp1a* and *foxp1b* mutants.

Effect on mean LD diameter (μm) is plotted per 200 μm stratum for *foxp1a*^{ed116} (left) and *foxp1b*^{ed125} (right) experiments. Four contrasts are shown: (a) diet effect (HFD minus control) in wild-type siblings (black); (b) diet effect in mutants (dark grey); (c) genotype effect (mutant minus wild-type) under control diet (medium grey); (d) genotype effect under high-fat diet (light grey). Dashed horizontal line indicates zero effect. Filled circles denote strata where the effect was statistically significant (Benjamini-Hochberg adjusted $p < 0.05$).

Data availability

Data, code and interactive HTML analysis files are available as a reusable pipeline at https://github.com/jeminchin/zebrafish_adipose_morphology_screen.

Acknowledgements

We thank Rosalyn Fong for help with initial zebrafish *foxp1* genotyping strategies, and David Duneau, Will Cawthorn and Rob Semple for advice on modelling of adipose morphology. The work was funded by a British Heart Foundation PhD Studentship awarded to RW, and a BBSRC grant (BB/X009467/1) awarded to PT and JENM.

Additional information

Author contributions

RW and PT performed experiments, RW, PT and JENM performed data analysis, PT and JENM wrote the manuscript.

Funding

Funder	Grant reference number	Author
UKRI Biotechnology and Biological Sciences Research Council (AFRC)	BB/X009467/1	Panna Tandon James Minchin
British Heart Foundation (BHF)	REA PhD Studentship	Rebecca Wafer

Additional files

[Supplemental Table 1](#)

[Supplemental Table 2](#)

[Supplemental Table 3](#)

[Supplemental Table 4](#)

References

1. Tandon P, Wafer R, Minchin JEN (2018) Adipose morphology and metabolic disease. *J Exp Biol* **221**:jeb164970
2. Hagberg CE, Spalding KL (2024) White adipocyte dysfunction and obesity-associated pathologies in humans. *Nat Rev Mol Cell Biol* **25**:270-89
3. Vishvanath L, Gupta RK (2019) Contribution of adipogenesis to healthy adipose tissue expansion in obesity. *J Clin Invest* **129**:4022-31
4. Rydén M, Arner P (2017) Cardiovascular risk score is linked to subcutaneous adipocyte size and lipid metabolism. *J Intern Med* **282**:220-8
5. Acosta JR, Douagi I, Andersson DP, Bäckdahl J, Rydén M, Arner P, et al. (2016) Increased fat cell size: a major phenotype of subcutaneous white adipose tissue in non-obese individuals with type 2 diabetes. *Diabetologia* **59**:560-70
6. Weyer C, Foley JE, Bogardus C, Tataranni PA, Pratley RE (2000) Enlarged subcutaneous abdominal adipocyte size, but not obesity itself, predicts type II diabetes independent of insulin resistance. *Diabetologia* **43**:1498-506
7. Hoffstedt J, Arner E, Wahrenberg H, Andersson DP, Qvisth V, Löfgren P, et al. (2010) Regional impact of adipose tissue morphology on the metabolic profile in morbid obesity. *Diabetologia* **53**:2496-503

8. **Boden G**, Cheung P, Mozzoli M, Fried SK (2003) Effect of thiazolidinediones on glucose and fatty acid metabolism in patients with type 2 diabetes. *Metabolism* **52**:753-9
9. **Kim SM**, Lun M, Wang M, Senyo SE, Guillermier C, Patwari P, et al. (2014) Loss of white adipose hyperplastic potential is associated with enhanced susceptibility to insulin resistance. *Cell Metab* **20**:1049-58
10. **Gao Z**, Daquinag AC, Fussell C, Zhao Z, Dai Y, Rivera A, et al. (2020) Age-associated telomere attrition in adipocyte progenitors predisposes to metabolic disease. *Nat Metab* **2**:1482-97
11. **Muir LA**, Neeley CK, Meyer KA, Baker NA, Brosius AM, Washabaugh AR, et al. (2016) Adipose tissue fibrosis, hypertrophy, and hyperplasia: Correlations with diabetes in human obesity. *Obesity* **24**:597-605
12. **Arner P**, Arner E, Hammarstedt A, Smith U (2011) Genetic predisposition for Type 2 diabetes, but not for overweight/obesity, is associated with a restricted adipogenesis. *PLoS One* **6**:e18284
13. **Flynn EJ**, Trent CM, Rawls JF (2009) Ontogeny and nutritional control of adipogenesis in zebrafish (*Danio rerio*). *J Lipid Res* **50**:1641-52
14. **Imrie D**, Sadler KC (2010) White adipose tissue development in zebrafish is regulated by both developmental time and fish size. *Dev Dyn* **239**:3013-23
15. **Morocho-Jaramillo P-A**, Kotlar-Goldaper I, Zakarauskas-Seth BI, Purfürst B, Filosa A, Sawamiphak S (2024) The zebrafish heart harbors a thermogenic beige fat depot analog of human epicardial adipose tissue. *Cell Rep* **43**:113955
16. **Minchin JEN**, Rawls JF (2017) A classification system for zebrafish adipose tissues. *Dis Model Mech* **10**:797-809
17. **Minchin JEN**, Scahill CM, Staudt N, Busch-Nentwich EM, Rawls JF (2018) Deep phenotyping in zebrafish reveals genetic and diet-induced adiposity changes that may inform disease risk. *J Lipid Res* **59**:1536-45
18. **Oka T**, Nishimura Y, Zang L, Hirano M, Shimada Y, Wang Z, et al. (2010) Diet-induced obesity in zebrafish shares common pathophysiological pathways with mammalian obesity. *BMC Physiol* **10**:21
19. **Landgraf K**, Schuster S, Meusel A, Garten A, Riemer T, Schleinitz D, et al. (2017) Short-term overfeeding of zebrafish with normal or high-fat diet as a model for the development of metabolically healthy versus unhealthy obesity. *BMC Physiol* **17**:4
20. **Minchin JEN**, Rawls JF (2011) In vivo analysis of white adipose tissue in zebrafish. *Methods Cell Biol* **105**:63-86
21. **Lumaquin D**, Johns E, Montal E, Weiss JM, Ola D, Abuhashem A, et al. (2021) An in vivo reporter for tracking lipid droplet dynamics in transparent zebrafish. *eLife* **10** PubMed
22. **Wilson MH**, Ekker SC, Farber SA (2021) Imaging cytoplasmic lipid droplets in vivo with fluorescent perilipin 2 and perilipin 3 knock-in zebrafish. *eLife* **10** PubMed
23. **Lepanto P**, Levin-Ferreyra F, Koziol U, Malacrida L, Badano JL (2021) Insights into in vivo adipocyte differentiation through cell-specific labeling in zebrafish. *Biol Open* **10**
<https://doi.org/10.1242/bio.058734>
24. **Mao Y**, Hong K-H, Liao W, Li L, Kim S-J, Xiong Y, et al. (2021) Generation of a novel transgenic zebrafish for studying adipocyte development and metabolic control. *Int J Mol Sci* **22**:3994
25. **Mazzaferro E**, Mujica E, Zhang H, Emmanouilidou A, Jenseit A, Evcimen B, et al. (2025) Functionally characterizing obesity-susceptibility genes using CRISPR/Cas9, in vivo imaging and deep learning. *Sci Rep* **15**:5408
26. **van der Klaauw AA**, Croizier S, de Oliveira E Mendes, Stadler LKJ, Park S, Kong Y, et al. (2019) Human Semaphorin 3 variants link melanocortin circuit development and energy balance. *Cell* **176**:729-742.
27. **Honecker J**, Ruschke S, Seeliger C, Laber S, Strobel S, Pröll P, et al. (2022) Transcriptome and fatty-acid signatures of adipocyte hypertrophy and its non-invasive MR-based characterization in human adipose tissue. *EBioMedicine* **79**

28. **Emont MP**, Jacobs C, Essene AL, Pant D, Tenen D, Colleluori G, et al. (2022) A single-cell atlas of human and mouse white adipose tissue. *Nature* **603**:926-33
29. **Civelek M**, Wu Y, Pan C, Raulerson CK, Ko A, He A, et al. (2017) Genetic Regulation of Adipose Gene Expression and Cardio-Metabolic Traits. *Am J Hum Genet* **100**:428-43
30. **Kroll F**, Powell GT, Ghosh M, Gestri G, Antinucci P, Hearn TJ, et al. (2021) A simple and effective F0 knockout method for rapid screening of behaviour and other complex phenotypes. *eLife* **10** <https://doi.org/10.7554/eLife.59683>
31. **Varshney GK**, Pei W, LaFave MC, Idol J, Xu L, Gallardo V, et al. (2015) High-throughput gene targeting and phenotyping in zebrafish using CRISPR/Cas9. *Genome Res* **25**:1030-42
32. **Wu RS**, Lam II, Clay H, Duong DN, Deo RC, Coughlin SR (2018) A Rapid Method for Directed Gene Knockout for Screening in G0 Zebrafish. *Dev Cell* **46**:112-125.
33. **Stringer C**, Wang T, Michaelos M, Pachitariu M (2021) Cellpose: a generalist algorithm for cellular segmentation. *Nat Methods* **18**:100-6
34. **Tandon P**, Lyall Z, Cowie M, Minchin JEN (2025) Human genetic studies and zebrafish models identify *Plxn4* as a regulator of adiposity, somatic growth, and feeding behaviours. *bioRxiv* <https://doi.org/10.1101/2025.03.15.643290>
35. **Arner E**, Westermark PO, Spalding KL, Britton T, Rydén M, Frisén J, et al. (2010) Adipocyte turnover: relevance to human adipose tissue morphology. *Diabetes* **59**:105-9
36. **Gustafson B**, Nerstedt A, Smith U (2019) Reduced subcutaneous adipogenesis in human hypertrophic obesity is linked to senescent precursor cells. *Nat Commun* **10**:2757
37. **Hedbacker K**, Lu Y-H, Dallner O, Li Z, Fayzikhodjaeva G, Birsoy K, et al. (2020) Limitation of adipose tissue by the number of embryonic progenitor cells. *eLife* **9** <https://doi.org/10.7554/eLife.53074>
38. **Loureiro Z Yang**, Joyce S, DeSouza T, Solivan-Rivera J, Desai A, Skritakis P, et al. (2023) Wnt signaling preserves progenitor cell multipotency during adipose tissue development. *Nat Metab* **5**:1014-28
39. **McMenamin SK**, Minchin JEN, Gordon TN, Rawls JF, Parichy DM (2013) Dwarfism and increased adiposity in the *gh1* mutant zebrafish *vizzini*. *Endocrinology* **154**:1476-87
40. **Zhao J**, Li H, Zhou R, Ma G, Dekker JD, Tucker HO, et al. (2015) *Foxp1* regulates the proliferation of hair follicle stem cells in response to oxidative stress during hair cycling. *PLoS One* **10**:e0131674
41. **Naudin C**, Hattabi A, Michelet F, Miri-Nezhad A, Benyoucef A, Pflumio F, et al. (2017) PUMILIO/FOXP1 signaling drives expansion of hematopoietic stem/progenitor and leukemia cells. *Blood* **129**:2493-506
42. **Li H**, Liu P, Xu S, Li Y, Dekker JD, Li B, et al. (2017) FOXP1 controls mesenchymal stem cell commitment and senescence during skeletal aging. *J Clin Invest* **127**:1241-53
43. **Pearson CA**, Moore DM, Tucker HO, Dekker JD, Hu H, Miquelajáuregui A, et al. (2020) *Foxp1* regulates neural stem cell self-renewal and bias toward deep layer cortical fates. *Cell Rep* **30**:1964-1981.
44. **Fu NY**, Pal B, Chen Y, Jackling FC, Milevskiy M, Vaillant F, et al. (2018) *Foxp1* Is Indispensable for Ductal Morphogenesis and Controls the Exit of Mammary Stem Cells from Quiescence. *Dev Cell* **47**:629-644.
45. **Liu P**, Huang S, Ling S, Xu S, Wang F, Zhang W, et al. (2019) *Foxp1* controls brown/beige adipocyte differentiation and thermogenesis through regulating β 3-AR desensitization. *Nat Commun* **10**:5070
46. **Dai S**, Qu L, Li J, Chen Y (2021) Toward a mechanistic understanding of DNA binding by forkhead transcription factors and its perturbation by pathogenic mutations. *Nucleic Acids Res* **49**:10235-49
47. **Stroud JC**, Wu Y, Bates DL, Han A, Nowick K, Paabo S, et al. (2006) Structure of the forkhead domain of FOXP2 bound to DNA. *Structure* **14**:159-66
48. **Glastonbury CA**, Pulit SL, Honecker J, Censin JC, Laber S, Yaghootkar H, et al. (2020) Machine Learning based histology phenotyping to investigate the epidemiologic and genetic basis of adipocyte morphology and cardiometabolic traits. *PLoS Comput Biol* **16**:e1008044

49. Kulyté A, Aman A, Strawbridge RJ, Arner P, Dahlman IA (2022) Genome-wide association study identifies genetic loci associated with fat cell number and overlap with genetic risk loci for type 2 diabetes. *Diabetes* **71**:1350-62
50. Ye RZ, Richard G, Gévry N, Tchernof A, Carpentier AC (2022) Fat cell size: Measurement methods, pathophysiological origins, and relationships with metabolic dysregulations. *Endocr Rev* **43**:35-60
51. Jeffery E, Wing A, Holtrup B, Sebo Z, Kaplan JL, Saavedra-Peña R, et al. (2016) The adipose tissue microenvironment regulates depot-specific adipogenesis in obesity. *Cell Metab* **24**:142-50
52. Minchin JEN, Dahlman I, Harvey CJ, Mejhert N, Singh MK, Epstein JA, et al. (2015) Plexin D1 determines body fat distribution by regulating the type V collagen microenvironment in visceral adipose tissue. *Proc Natl Acad Sci U S A* **112**:4363-8
53. Chau Y-Y, Bandiera R, Serrels A, Martínez-Estrada OM, Qing W, Lee M, et al. (2014) Visceral and subcutaneous fat have different origins and evidence supports a mesothelial source. *Nat Cell Biol* **16**:367-75
54. Choi E-H, Park S-J (2023) TXNIP: A key protein in the cellular stress response pathway and a potential therapeutic target. *Exp Mol Med* **55**:1348-56
55. Russell-Guzmán J, Américo-Da Silva L, Cadagan C, Maturana M, Palomero J, Estrada M, et al. (2024) Activation of the ROS/TXNIP/NLRP3 pathway disrupts insulin-dependent glucose uptake in skeletal muscle of insulin-resistant obese mice. *Free Radic Biol Med* **222**:187-98
56. Hamilton JP, Potter JJ, Koganti L, Meltzer SJ, Mezey E (2014) Effects of vitamin D3 stimulation of thioredoxin-interacting protein in hepatocellular carcinoma: TXNIP, vitamin D3 and HCC. *Hepatol Res* **44**:1357-66
57. Freeburg SH, Shwartz A, Kemény LV, Smith CJ, Weeks O, Miller BM, et al. (2024) Hepatocyte vitamin D receptor functions as a nutrient sensor that regulates energy storage and tissue growth in zebrafish. *Cell Rep* **43**:114393
58. Chun T-H, Hotary KB, Sabeh F, Saltiel AR, Allen ED, Weiss SJ (2006) A pericellular collagenase directs the 3-dimensional development of white adipose tissue. *Cell* **125**:577-91
59. Li X, Zhao Y, Chen C, Yang L, Lee H-H, Wang Z, et al. (2020) Critical role of matrix metalloproteinase 14 in adipose tissue remodeling during obesity. *Mol Cell Biol* **40**:1-24
60. Kirstein AS, Kehr S, Nebe M, Hanschkow M, Barth LAG, Lorenz J, et al. (2021) PTEN regulates adipose progenitor cell growth, differentiation, and replicative aging. *J Biol Chem* **297**:100968
61. Huang W, Queen NJ, McMurphy TB, Ali S, Cao L (2019) Adipose PTEN regulates adult adipose tissue homeostasis and redistribution via a PTEN-leptin-sympathetic loop. *Mol Metab* **30**:48-60
62. Cereijo R, Quesada-López T, Gavalda-Navarro A, Tarascó J, Pellitero S, Reyes M, et al. (2021) The chemokine CXCL14 is negatively associated with obesity and concomitant type-2 diabetes in humans. *Int J Obes* **45**:706-10
63. Zachara M, Rainer PY, Hashimi H, Russeil JM, Alpern D, Ferrero R, et al. (2022) Mammalian adipogenesis regulator (Areg) cells use retinoic acid signalling to be non- and anti-adipogenic in age-dependent manner. *EMBO J* **41**:e108206
64. Ampudia-Mesias E, El-Hadad S, Cameron CS, Wöhrer A, Ströbel T, Saydam N, et al. (2022) SRPX emerges as a potential tumor marker in the extracellular vesicles of glioblastoma. *Cancers* **14**:1984
65. Zou Y, Gong N, Cui Y, Wang X, Cui A, Chen Q, et al. (2015) Forkhead box P1 (FOXP1) transcription factor regulates hepatic glucose homeostasis. *J Biol Chem* **290**:30607-15
66. Ma H, Sukonina V, Zhang W, Meng F, Subhash S, Palmgren H, et al. (2023) The transcription factor Foxp1 regulates aerobic glycolysis in adipocytes and myocytes. *J Biol Chem* 104795

Peer reviews

Reviewer #1 (Public review):

In this manuscript, Wafer and Tandon et al. present a thoughtful and well-designed genetic screen for regulators of adipose remodeling using zebrafish as a model system. The authors cross-referenced several human adipocyte-related transcriptomic and genetic association datasets to identify candidate genes, which they then functionally tested in zebrafish. Importantly, the authors devised an unbiased microscopy-based screening platform to document quantitative adipose phenotypes with whole animal imaging, while also employing rigorous statistical methods. From their screen, the authors identified 3 genes that resulted in robust adipose phenotypes out of a total of 25 that were tested. Overall, this work will be an important resource for the field because of the genes identified from the screen, the quantitative screening pipeline, and the rigorous phenotypic analysis.

Comments on revisions:

The authors have far exceeded my expectations with their revised manuscript. All my questions and concerns from the original manuscript have been addressed by the authors. The additional data and analysis in Figure 6 and Supplementary Figure 8 are compelling and have greatly improved the manuscript.

<https://doi.org/10.7554/eLife.107327.2.sa2>

Reviewer #2 (Public review):

This manuscript by Wafer, Tandon et al., presents exciting new approaches for using the zebrafish CRISPR screening and imaging system to identify genes that are associated with hyperplastic and hypertrophic adipose morphology. This paper established valuable screening pipelines in zebrafish to identify genetic regulators that affect adipose tissue morphology by combining CRISPR with an imaging-based, comprehensive adipose spatial analysis platform. Starting from a human transcriptomic dataset with differentially expressed genes that separate small and large adipocytes, they eventually identified 3 genes that induce hyperplastic or hypertrophic phenotypes in zebrafish. From which, they focused on *foxp1* gene, a transcription factor known to regulate tissue development. They discovered that the *foxp1* mutant displays basal hypertrophic morphology and failed to undergo hypertrophic remodeling in response to a high-fat diet, suggesting a link between adipose tissue development and diet-induced remodeling response. Overall, this manuscript is extremely well-written, the data presented is quite compelling, and the identified novel genes that are associated with adipose tissue hyperplastic and hypertrophic morphology and diet-induced remodeling are very exciting.

Strength:

(1) Obesity remains a worldwide public health concern. The mechanisms underlying adipose tissue hypertrophic and hyperplastic adaptation remain unclear.

(2) This manuscript combined multiple omic datasets to identify candidate genes and performed a CRISPR-based screening to identify genes underlying adipose tissue development and adaptation. This new method will open opportunities that will facilitate our understanding and testing of new genetic mechanisms underlying the development of obesity.

(3) Using the screening approach, this paper successfully identified new genes that are associated with adipose tissue LD size change. More importantly, the paper provided further

validation using a stable CRISPR line to show the phenotype in basal and HFD conditions.

(4) The experiments are extremely well-designed. Sample sizes are large. Statistical analysis is rigorous. Overall, this is a very high-quality study.

Author's response to the previous comments/weakness:

(1) In this revised manuscript, the authors provided new comprehensive spatial analyses of *foxp1a* and *foxp1b* mutants in basal conditions as well as responding to high-fat feeding. The new data confirmed their initial findings and beautifully illustrated the spatiotemporal dynamics of the adipocytes in response to High-fat diet feeding.

(2) The authors have addressed all my comments, and I do not have further comments.

<https://doi.org/10.7554/eLife.107327.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Thank you for the thoughtful and constructive comments on our manuscript. We have carefully addressed all points raised, and believe the manuscript is substantially improved as a result. In particular, we have performed:

- Comprehensive spatial analysis of stable mutants. Following Recommendations for the authors comment #1, we performed spatial analysis by binning the anterior-posterior axis into 200 μ m strata. This analysis validates our initial conclusions and reveals striking spatiotemporal dynamics, including profoundly blunted HFD responses in *foxp1b* mutants (68% reduction) and loss of spatial gradients in *foxp1a* mutants.

- Substantially enhanced the statistical rigour of the screen analysis. We have implemented stratified Kolmogorov-Smirnov tests (within-experiment testing, then combined via Fisher's method) alongside linear mixed models to control for batch effects. In the revised manuscript, we now focus on three hypertrophy genes – *foxp1b*, *txnipa* and *mmp14b* – which are robustly validated by both methods.

- Normalisation of adipose area to body size. To address concerns about developmental delay (Recommendations for the authors #2), we now normalise adipose area to standard length. With this normalisation, *foxp1b* single mutants show only a non-significant trend toward decreased adiposity (updated from our original analysis), while the hypertrophic LD morphology remains highly significant - demonstrating the phenotype is independent of body size and not a developmental delay.

- Revised title. As suggested by Recommendations for the authors comment #6, we have changed the title to: "A quantitative in vivo CRISPR-imaging platform identifies regulators of hyperplastic and hypertrophic adipose morphology in zebrafish"

- Extensive code and analysis availability. We now provide all code and extensive analysis pipelines in interactive HTML documents at

https://github.com/jeminchin/zebrafish_adipose_morphology_screen 

Joint Public Review:

We thank the reviewers for their thoughtful assessment of our work and their recognition of the rigorous experimental design, statistical approaches, and the utility of both the identified genes and screening pipeline for the field. We address their concerns below.

Weakness:

Distinguishing developmental patterning from adipose tissue plasticity

We appreciate this important distinction and agree that separating developmental from adaptive effects is a key challenge in the field. We would like to make several points in response:

First, we acknowledge this limitation in our discussion and have now expanded this section to more explicitly address the interpretive boundaries of our approach. Our screening platform was intentionally designed to capture the outcome of genetic perturbation across development and early adaptation, as these processes are inherently intertwined during the establishment of adipose tissue.

Second, regarding the suggested analysis of lipid droplet size along the AP axis in response to HFD: we have now performed this analysis and include it as new Fig. 6 and new Supplemental Fig. 8 & 9. These data validate our initial conclusions and reveal striking spatiotemporal dynamics, including profoundly blunted HFD responses in *foxp1b* mutants (68% reduction) and loss of spatial gradients in *foxp1a* mutants. Further, these data provide additional resolution on regional responses to dietary challenge.

Third, we note that our stable mutant validation experiments (Figure 6) do begin to disentangle these effects by examining both baseline and HFD-challenged conditions in animals with constitutive genetic loss. However, we agree that definitive separation would require temporally controlled genetic manipulation, which we now acknowledge as an important future direction.

Lack of tissue-specific manipulations

We agree that tissue-specific approaches would strengthen mechanistic conclusions and have acknowledged this limitation in our revised discussion. The current study was designed as a discovery-focused screen to identify candidate regulators, with the understanding that mechanistic dissection would require follow-up studies employing tissue-specific tools.

We note that adipocyte-specific Cre/lox or Gal4-UAS approaches in zebrafish are feasible and represent an important next phase of investigation for the most promising candidates identified here, rather than a requirement for the current screening study. We have added text explicitly framing our findings as establishing genetic associations that warrant future tissue-autonomous investigation.

Recommendations for the authors:

(1) Analysis: In Figure 6, the authors state that *foxp1b* mutants "fail to undergo further hypertrophic remodeling in response to a high-fat diet (HFD)." *Foxp1b* mutant juveniles are already hypertrophic before the high-fat diet. After a high-fat diet, these mutants reach mean lipid droplet diameters similar to WT, approximately 65 μm , which the authors state earlier in the manuscript are "a potential upper limit of LD growth at this developmental stage." The authors should perform additional analysis of their existing data. Specifically, determine lipid droplet size by binning the AP axis as shown in Figure 3. The rationale is that lipid droplet size differences in response to HFD may be more evident when not considering the anterior populations of lipid droplets that have already reached maximum steady state size for this juvenile stage. This would not require any new experiments, just reanalyzing data similar to how they did in Figure 3.

We thank the reviewer for this excellent suggestion. We have performed the requested spatial analysis by binning the AP axis into 200 μm strata (Figure 3 approach). These data can be found in new Fig. 6H-M, and new Supplemental Figs 8 & 9. This new analysis verifies our initial conclusions, and also reveals several very interesting spatiotemporal dynamics

(i) Baseline hypertrophy in *foxp1b* mutants across AP strata

In support of our initial conclusion that *foxp1b* mutants have larger LDs at baseline, the spatial analysis confirms that on a control diet (baseline), *foxp1b* mutants have significantly larger LDs than WT across strata 1-5 (new Fig. 6I), ranging from +22.2 μm larger in strata 1 to +17.8 μm larger in strata 5 (all FDR-adjusted $p < 0.05$, linear mixed effects model). Extended analysis across all 15 strata is shown in Supplemental Figs. 8 & 9. By contrast, and also in support of our initial conclusion, *foxp1a* mutants showed no baseline hypertrophy on control diet (all strata $p > 0.10$, Supplemental Fig. 8).

(ii) *foxp1b* mutants show a profoundly blunted hypertrophic response to HFD

Using paired analysis (same fish on both control diet and after 14 days of high-fat diet) with a linear mixed effects model, we quantified the effect of HFD across all strata:

(A) Anterior/oldest strata (1-6): WT + HFD increases LD diameter by +25.1-28.1 μm (+52-58%, $p < 0.0001$). Whereas, *foxp1b* mutants + HFD only increase LD diameter by +7.5-11.7 μm (+12-19%, $p < 0.003$). Therefore, in the oldest/most anterior regions, containing the largest LDs, the hypertrophic response of *foxp1b* mutants to HFD is ~57% weaker than WTs.

(B) Posterior/newer strata (7-15): WT + HFD undergo significant increases in LD diameter of +17.7-23.7 μm ($p < 0.024$). However, in *foxp1b* mutants there is no significant hypertrophic response at all ($p > 0.068$), and hypertrophic effect sizes decline from +6.8 μm (stratum 7) to +0.4 μm (stratum 15).

(C) Overall effect: Averaged across all strata, WT + HFD LDs show +24.4 μm increase ($p < 0.0001$), whereas *foxp1b* mutant LDs only show a +7.7 μm increase with HFD ($p = 0.020$). Therefore, *foxp1b* mutants show a 68% reduction in hypertrophic growth in response to HFD compared to WT (Fig. 6K).

The consequence of these spatial dynamics is that WT SAT LDs - which start 22 μm smaller than *foxp1b* mutants on a control diet - undergo massive hypertrophy across all regions/strata in response to a HFD. Meanwhile, *foxp1b* mutants - starting larger than in WTs - show only a modest, spatially restricted response. This results in a convergence in LD size in early/anterior strata, but WT LDs actually surpass *foxp1b* mutant sizes in late/posterior strata (strata 14-15: +WT 14.7 μm larger on HFD, $p = 0.028$; Supplemental Figs. 8 & 9).

By contrast, *foxp1a* mutants retain the capacity for HFD-induced hypertrophy but show a ~35% weaker response than WT ($p = 0.023$) – significantly less severe than the 68% reduction in *foxp1b* mutants. Interestingly, *foxp1a* mutants after HFD show a reduction in the AP gradation of LD size observed in WT and *foxp1b* mutants (uniform +14.4 mm across all strata versus WT range of +26.4 mm anteriorly to +16.6 mm posteriorly), suggesting that *foxp1a* may regulate spatial heterogeneity in adaptive responses to HFD (Fig. 6L-M).

(iii) Developmental ceiling or impaired adaptive capacity?

The reviewer raises an important question about whether anterior adipose LDs have reached a "developmental ceiling." After conducting the spatial analysis suggested by the Reviewer, we now believe several lines of evidence support an intrinsic defect in HFD-induced hypertrophy in *foxp1b* mutants, rather than reaching a developmentally determined limit:

First, *foxp1b* mutants show reduced responses across ALL strata, not just anterior regions. The attenuation extends throughout the entire AP axis (57% reduction in strata 1-6, complete loss of response in strata 7-15). If anterior adipocytes had simply reached a size ceiling, we would expect normal responses in posterior regions where cells are smaller - but we don't observe this.

Second, in posterior/newer regions of SAT (strata 14-15) the hypertrophic response to HFD in *foxp1b* is so limited that WT LDs actually become larger than *foxp1b* mutant LDs (+14.7 mm larger, $p = 0.028$; Supplemental Fig. 9). This demonstrates that these LD sizes are not developmentally limiting and argues for intrinsic hypertrophic defects in response to HFD.

Third, *foxp1a* mutants provide an important control. These mutants show no baseline hypertrophy (all strata $p > 0.10$) yet still exhibit blunted hypertrophic responses to HFD (~35% reduction, $p = 0.023$), proving that reduced HFD responses can occur independently of baseline hypertrophy.

We have updated the Results and Discussion to reflect these new conclusions. Methods have been updated to include the spatial analysis approach.

(2) Adipose morphogenesis in WT is a function of standard length, as shown by the authors. At juvenile stages, foxp1 mutants are both smaller and have reduced adipocyte coverage, while adults show normal body length and very subtle adipose phenotypes. Can the authors demonstrate that the observed defects in foxp1 mutant juveniles are bona fide phenotypes rather than a developmental delay?

We thank the reviewer for this key point. We agree it is critical to distinguish true *foxp1b*-dependent phenotypes from potential developmental delay. Importantly, our data strongly argue against a simple developmental delay. We show that LD size scales with body size in Fig. 3G, with smaller zebrafish having smaller LDs and larger zebrafish having larger LDs. In contrast to a developmental delay, our data show that *foxp1b* single and *foxp1a;foxp1b* double mutants are smaller (reduced standard length) but have larger LDs (Fig. 6E,G). This dissociation between body size and LD size is the opposite of what would be expected from developmental delay.

To account for the body size difference, we have now normalised adipose area to standard length (Fig. 6F). With this normalisation, *foxp1b* single mutants show only a non-significant trend toward decreased adiposity, whereas *foxp1a;foxp1b* double mutants remain significantly reduced. This represents a change from our original analysis and we have updated the text accordingly. Critically, despite normalised adipose area showing only a trend in *foxp1b* singles, the hypertrophic LD morphology remains highly significant (Fig. 6G), demonstrating that the morphological phenotype is robust and independent of overall body size.

We have clarified this interpretation in the Results and Discussion.

(3) What was the rationale for selecting one amongst paralogous genes for the screen? For example, why did the authors choose ptenb rather than ptena?

(4) Point 3 is particularly relevant for the final six genes that resulted in adipose phenotypes. Why did the authors choose not to target both paralogs, given that multiplexed F0 CRISPR targeting is feasible in zebrafish (PMID: 29974860).

We answer Points 3 & 4 together here.

We used the DIOPT (DRSC Integrative Ortholog Prediction Tool) orthology tool to identify the zebrafish paralogue with the highest orthology score to each human gene. This tool integrates predictions from 20 orthology databases to generate a composite score. We selected the paralogue with the highest DIOPT score for each gene. For example, we selected *ptenb* over *ptena* because it showed a higher predicted orthology to human PTEN.

We acknowledge this approach has important limitations, including orthology scores not necessarily predicting functional equivalence (ie, the "most orthologous" paralogue may not be the one with the most relevant adipose tissue function in zebrafish). We acknowledge that

this may mean we have missed genuine hits - testing only one paralogue means we could fail to identify genes where the "less orthologous" paralogue has the relevant adipose function.

Our findings with *Foxp1* paralogues both validate this approach and reveal its limitations. The higher-scoring paralogue *foxp1b* (DIOPT score = 13/19) showed the more severe phenotype, validating our prioritisation. However, the lower-scoring paralogue *foxp1a* (DIOPT score = 5/19), which we tested subsequently, showed a distinct but significant phenotype (altered spatial patterning) – a finding that would have been missed had we not pursued secondary validation.

For future screens where comprehensive hit identification is the goal, multiplexed targeting of all paralogues would be valuable, though this may complicate interpretation of paralogue-specific phenotypes. We have discussed this in the Discussion.

(5) General framework and limitations: The analysis platform presented in the manuscript cannot separate the developmental effects from adipose tissue plasticity/remodeling. Potential approaches that may help address this concern include: (a) establishing a baseline model to illustrate how WT fish respond to high-fat diet (HFD); (b) showing how mutants with hyperplasticity (opposite effects of foxp1 mutants) respond to HFD; (c) examining whether foxp1 gene expression level changes in response to HFD. However, these approaches (especially a and b) would require extensive experimental work and may be beyond the scope of this study. Without further evidence or data support of adipose tissue plasticity and remodeling, the author may want to emphasize in the background and discussion sections how adipose tissue development may affect plasticity and adaptation, and soften the tone of how genes may directly regulate adipose tissue plasticity and adaptation.

We thank the reviewer for this comment about the relationship between adipose development and plasticity/remodelling. We agree this is an important issue as we are looking in juvenile fish that are still growing. Therefore, when we feed them HFD and see LDs get bigger – is this diet-induced remodelling or just accelerated normal development (ie, growth that would happen anyway, but occurring faster due to more nutrients)?

To address the reviewer's specific suggestions:

(A) Baseline model of WT HFD response: We have now performed detailed spatial analysis of WT responses to HFD (new Fig 6H-M, Supplemental Figs. 8 & 9). This analysis establishes a comprehensive baseline for hypertrophic responses to HFD in developing adipose tissue. In summary, WT fish show robust, statistically significant and spatially-graded hypertrophic responses to HFD across the entire AP axis, with responses ranging from +28.1 mm anteriorly to +17.7 mm posteriorly.

We agree with the Reviewer that separating developmental from adaptive processes in growing juvenile fish is challenging. Importantly, we believe *foxp1a* mutants provide compelling genetic evidence that we are studying adaptive responses rather than purely developmental processes. *foxp1a* mutants have normal baseline LD sizes on control diet (demonstrating *foxp1a* is not required for developmental adipose expansion), yet when challenged with HFD show significantly reduced hypertrophic expansion and reduction of spatial gradient. This genetic dissociation strongly argues we are observing adaptive capacity rather than developmental growth rate.

(B) Hyperplastic mutants:

We agree that analysis of hyperplastic mutants would provide valuable complementary information about tissue remodelling capacity. However, as the reviewer anticipated, this would require: (1) generating stable lines of the appropriate hyperplastic mutants, (2) conducting paired HFD feeding studies, (3) performing spatial morphometric analysis

comparable to our *foxp1* studies, and (4) potentially distinguishing hyperplastic vs hypertrophic contributions to expansion. We agree this constitutes substantial additional experimental work beyond the scope of the current manuscript, though it represents an important direction for future studies.

(C) *foxp1* expression changes in HFD:

Unfortunately, we do not have SAT samples from HFD-treated fish preserved for RNA analysis, and therefore cannot assess whether *foxp1* expression levels change in response to dietary challenge. This would be valuable for future studies to determine whether *foxp1* genes are dynamically regulated during metabolic adaptation or function as constitutive regulators of adaptive capacity.

Following the Reviewer's guidance, we have revised throughout the manuscript to more carefully distinguish developmental patterning from metabolic adaptation.

(6) Title: In the absence of experimental results that can distinguish between developmental effects from adipose tissue plasticity/remodeling, such as those mentioned above, the manuscript title is not accurate and should therefore be revised to be something like "hyperplastic and hypertrophic adipose morphology."

We have now altered the title as the Reviewer suggested to “A quantitative in vivo CRISPR-imaging platform identifies regulators of hyperplastic and hypertrophic adipose morphology in zebrafish”

Minor:

(7) In mice studies, deleting foxp1b in adipose tissue protects mice from diet-induced obesity, while overexpressing foxp1b in adipose tissue promotes diet-induced obesity (Liu et al., Nature Communication, 2019). These overall phenotypes and foxp1b-mediated effects appear to be contradictory to what is observed in the zebrafish model. Can the authors also provide more evidence/discussion on why such a difference occurs comparing zebrafish and mice models?

We thank the reviewer for this important comparison. We believe the apparent contradictions reflect (1) differences in adipose tissue thermogenic capacity - between species possibly, but also between functionally distinct depots and (2) whole-organism versus tissue-specific experimental approaches.

(1) Different adipose tissue biology: browning-prone vs browning-resistant adipose

Liu et al. (2019, PMID: 31699980) demonstrated that adipose-specific deletion of *Foxp1* in mice increases thermogenesis and browning of SAT, with protection from diet-induced obesity (DIO) and improved insulin sensitivity. Conversely, *Foxp1* overexpression impaired adaptive thermogenesis and promoted DIO. Mechanistically, *Foxp1* directly represses β 3-adrenergic receptor transcription, thereby inhibiting the thermogenic program. Strikingly, mouse *Foxp1*-deleted adipocytes displayed smaller, multilocular lipid droplets characteristic of brown/beige adipocytes.

These morphological outcomes initially appear opposite to our zebrafish findings: mouse *Foxp1* mutants have smaller adipocytes (due to browning), while zebrafish *foxp1b* mutants have larger lipid droplets (hypertrophy). We believe this fundamental difference may reflect the propensity of adipose tissue to undergo adaptive thermogenesis.

While it was recently discovered that zebrafish possess thermogenic epicardial adipose tissue (PMID: 38507414), in general zebrafish adipose is not considered thermogenic, and zebrafish as ectotherms are thought to lack adaptive thermogenesis for thermoregulation. The exact thermogenic potential of zebrafish adipose remains to be fully characterised, but potential

differences in thermogenic capacity between mouse and zebrafish adipose may help explain the distinct phenotypic outcomes.

Importantly, Liu et al. studied mouse inguinal subcutaneous WAT - the depot most prone to browning in rodents. It remains unclear what role Foxp1 plays in browning-resistant mammalian WAT depots, where thermogenic conversion does not readily occur. In such depots, Foxp1 loss might produce phenotypes more similar to our zebrafish findings - dysregulated white adipose function without browning.

The above hypothesis suggest that browning responses may mask other roles for Foxp1 in WAT. Interestingly, although not quantified in the paper, Liu et al.'s Foxp1 overexpression model (Ap2-Foxp1) appeared to reduce adipocyte size despite suppressing Ucp1 expression and reducing lipolysis. These data suggest more complex roles and indicate that Foxp1's control of adipocyte size might extend beyond simply regulating thermogenesis and may involve coordinating the balance between hyperplastic versus hypertrophic expansion.

Furthermore, human subcutaneous WAT is not as prone to browning as mouse inguinal WAT. Human browning occurs primarily in specialised depots (e.g. supraclavicular, deep neck), while the majority of human adipose tissue represents constitutive white adipose with limited thermogenic capacity. Therefore, it remains an open question whether FOXP1's primary physiological role in humans relates to thermogenesis regulation (in specialised depots) or white adipose metabolic control (in the majority of adipose tissue). Zebrafish findings examining constitutive WAT function (admittedly the lack of adaptive thermogenesis in zebrafish is presumed at this stage) may be more relevant to human adipose than initially appear.

(2) Whole-organism vs tissue-specific effects on metabolic health

A second apparent contradiction concerns metabolic outcomes: mouse adipose-specific Foxp1 deletion improves metabolic health (Liu et al.), whereas our zebrafish whole-organism *foxp1b* mutants display metabolic dysfunction (baseline hypertrophy, impaired HFD response, hyperglycaemia and fatty liver). We believe this discrepancy reflects comparison of whole-animal mutants (zebrafish) to tissue-specific deletions (mouse), rather than opposite adipose tissue functions.

Critically, Foxp1 has established roles in hepatic glucose metabolism. Zou et al. (PMID: 26504089) demonstrated that hepatic Foxp1 inhibits expression of gluconeogenesis genes and decreases hepatic glucose production and fasting blood glucose by competing with Foxo1 for binding of insulin responsive gluconeogenic genes. In line with these observations, we observe fatty liver and hyperglycaemia in *foxp1a;foxp1b* double mutant zebrafish (data not shown), suggesting that the metabolic dysfunction in our whole-animal mutants may be driven primarily by hepatic Foxp1 loss rather than adipose-specific effects.

We have expanded on the points raised here in the Discussion.

(8) Line 522-524: "The major phenotype in *foxp1a* mutants was impaired adipose expansion following HFD, suggesting failure to respond to diet-induced stress signals". In the presented Figure 6j, *foxp1a* mutant expands adipose LD size following HFD, similar to the control, which is contradictory to the statement above. Please clarify.

We thank the reviewer for highlighting this apparent inconsistency and apologise for imprecise wording. These measurements are actually consistent but refer to different scales of analysis.

Tissue level (Supplementary Fig. 7): *foxp1a* mutants show significantly reduced total adipose expansion (based on whole-animal Nile Red images) compared to wild-type fish on HFD—this is what we refer to as "impaired adipose expansion."

Cellular level (Fig. 6L-M): At the individual adipocyte level, *foxp1a* mutants show statistically significant increases in LD diameter following HFD. However, the magnitude is reduced by ~35% compared to wild-type (mutants: +14.4 μm ; WT: +22.2 μm ; $p = 0.023$).

We have revised the text to more precisely state "reduced adipose expansion" rather than "impaired expansion" to avoid implying complete failure to respond.

<https://doi.org/10.7554/eLife.107327.2.sa0>